The Household Diary Study

Mail Use & Attitudes in FY 2005



Demand Forecasting and Economic Analysis,
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Executive Summary

This report documents the findings of the United States Postal Service's Household Diary Study (HDS) for Fiscal Year (FY) 2005. The three main study purposes are to:

- Measure the mail sent and received by U.S. households,
- Provide a means to track household mail trends over time, and
- Make comparisons of mail use between different types of households.

The report examines these trends in the context of changes and developments in the wider markets for communications and package delivery.

Background

The Household Diary Study survey, fielded continuously since 1987, aims to collect information on household use of the mail and how that use changes over time. The survey collects household information on demographics, lifestyle, attitudes toward mail and advertising, bill payment behavior, and use of the Internet and other information technologies.

The FY 2005 report covers Government Fiscal Year 2005, with comparisons to 2003 and 2004. Note: 2003 data published in this report differs from the original annual report due to a shift from Postal Fiscal Year to Government Fiscal Year.

The aim of the HDS is to collect information on household use of the mail and how that use changes over time.

Overview

Households received 151.6 billion pieces of mail in 2005, and sent 21.7 billion, as seen in Table E.1. Mail sent or received by households constituted 82 percent of total domestic mail in FY 2005. Forty-eight percent of the mail that households received was sent Standard Mail. Only about four percent of household mail (three percent of total mail) was sent between households; the rest was sent between households and nonhouseholds.

Table E.1:
Mail Sent and Received by Households
(Billions of Pieces)

Mail Classification	Received	Sent
First-Class:	58.8	21.1
Standard Regular Mail	70.1	—
Standard Nonprofit Mail	13.4	—
Periodicals	6.7	—
Packages	2.1	.3
Expedited	.5	.2
Total	**151.6**	**21.7**
Household-to-Household	6.2	
Total Mail Sent and Received by Households	167.0	
FY 2005 RPW Total	210.9	
Nonhousehold-to-Nonhousehold (Residual)	43.9	
Unaddressed	15.6	—

Source: Household Diary Study, FY 2005.
Note: Expedited includes Priority Mail and Express Mail.

Mail Markets

The Household Diary Study examines mail by the markets it serves. This design cuts across classes, but provides a foundation for understanding mail flows and the marketplace changes that affect them. Table E.2 shows the volume of household mail by market for 2003, 2004, and 2005.

Table E.2:
Household Mail Volume Sent and Received by Market Served
(Billions of Pieces)

Market	2003	2004	2005
Correspondence	16.9	17.2	16.6
Transactions	42.7	42.5	42.7
Advertising	92.1	94.8	101.6
Periodicals	6.8	6.7	6.7
Packages	2.6	2.6	2.4
Unclassified	5.0	4.5	4.5
Total	**158.3**	**160.8**	**167.0**

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Correspondence and Transactions include 7.5 billion pieces of First-Class advertising-enclosed mail (excluded from totals). Package Volumes include ground packages and expedited.

Around 35 percent of household mail contains correspondence and transactions, down from about 37 percent in 2003. In part, the decline in the percentage of correspondence is a continuation of long-term trends, but it is also related to changing demographics. Younger households both send and receive fewer pieces of correspondence mail, a result of younger households' comfort with and use of the Internet, and their lower average income level.

The increase in e-mail correspondence and the movement towards electronic bill payment have affected mail volumes.

Electronic alternatives also affect transactions mail volume. Automatic deduction and Internet bill payment account for a growing share of household bill payments over time. In fact, over the last five years, the percentage of bills paid by electronic methods increased from 10 percent to 24 percent. In contrast, bills paid by mail decreased from 77 percent to 67 percent of total payments. Similarly, payments in person decreased from 13 percent to 9 percent of all payments made. Also, the number of bills and statements households receive through the mail is declining on a per household basis while the number of bills received over the Internet is growing rapidly.

Advertising mail represented more than half of all mail received by households in 2005. As shown in Table E.3, about 82 percent of all advertising mail received by households is Standard Mail (83.5 billion pieces). The remainder consists of First-Class Mail; either stand-alone advertising (10.5 billion pieces), or advertising along with other matter (7.5 billion pieces). Over time, the data show a decline in the percentage of advertising that travels as First-Class mail.

Table E.3:
Advertising by Mail Class

Mail Classification	Volume *(Billions)*	Percent of Total Advertising
First-Class Advertising	18.1	18%
Standard Regular Mail	70.1	69%
Standard Nonprofit Mail	13.4	13%
Total Advertising Mail	**101.6**	**100%**

Source: Household Diary Study, FY 2005.

Households received 6.7 billion pieces of Periodicals mail in 2005, down about one percent from 2003. Three-quarters of these pieces were magazines.

Households received 2.4 billion packages in 2005 and sent 541 million packages. Packages sent are up slightly from 2003. Delivery from mail-order and Internet retailers is an important driver of package volume, and there are indications that increasing use of online auction sites such as eBay are responsible for some of the increase in the number of packages sent by households.

Table E.4:
Packages Sent and Received via the U.S. Postal Service
(Millions of Pieces)

Mail Classification	2005			
	Sent		Received	
	Number	Percent	Number	Percent
First-Class	195	36%	626	26%
Expedited	172	32%	477	20%
Standard Mail	—	—	802	34%
Package Services	121	22%	399	17%
Unclassified	54	10%	71	3%
Total Packages	**541**	**100%**	**2,375**	**100%**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding. Expedited includes Priority Mail and Express Mail.

Chapter 1: Introduction – Volumes & Trends

The U.S. Postal Service Household Diary Study (HDS) Report documents the findings of the Fiscal Year (FY) 2005 study. The HDS measures the mail sent and received by U.S. households, tracks household mail trends and compares mail use between different types of households.

The Household Diary Study provides a means to track household mail trends over time.

The Survey

The Household Diary Study survey, fielded continuously since 1987, aims to collect information on household use of the mail and how that use changes over time. The survey collects household information on:

- Demographics,
- Lifestyle,
- Attitudes toward mail and advertising,
- Bill payment behavior, and
- Use of the Internet and other information technologies.

These data are used for market research, forecasting, and strategic planning within the Postal Service.

The Survey Consists of Three Parts:

1) An **entry interview**, conducted by phone, collects demographic, lifestyle, and attitudinal information from about 8,500 households.
2) These households then receive a **mail diary**, which collects information on the mail the household sends and receives in a one-week period. Annually, about 5,200 households, or 100 each week, successfully complete the diary.
3) In addition, every individual over thirteen years old in a subset of these households is asked to complete a **personal technology questionnaire** on their use of and attitude toward technology.

The data generated by these three instruments are the basis of the analysis in this report.

The FY 2005 report covers the period from September 27, 2004, through September 30, 2005, equivalent to the Government Fiscal Year (GFY) used by the Postal Service (which runs from October 1 to September 30). Data from FY 2004 are also reported on a GFY basis. Data from earlier years are presented on a Postal Fiscal Year basis, consisting of 52-week periods that shifted slightly in relation to calendar dates.

U.S. Postal Service Volumes

Serving a nation containing five percent of the world's population, the Postal Service delivers almost 45 percent of the world's mail. The Postal Service delivered 212 billion pieces of mail in FY 2005 – an increase of 6 billion pieces from 2004. Standard Mail volume exceeded First-Class Mail for the first time in 2005. The 1.8 billion decrease in First-Class letters was a major contributor to the change in mix. In fact, since 1998, First-Class letters have decreased by 10.9 billion, or 20 percent, fueled, in part, by the gradual adoption of the Internet and the general increase in electronic diversion.

The Postal Service estimates the revenues, volumes, and weight of mail pieces going through the postal network using a combination of statistical sampling systems, mailing statements, and accounting data. These data are published in the Revenue, Pieces, and Weight (RPW) Reports. Table 1.1 presents the volumes from the report for FY 2005, along with data for FY 1987, 2003, and 2004. This report makes frequent comparisons to those years.

Table 1.2 reports revenue, pieces, and weight data by class and shape for FY 2005.

- **The letters** column heading includes postcards and refers to pieces that are less than 11.5 inches wide by 6.125 inches tall, and less than .25 inches thick.
- **Flats** consist of pieces that are greater than 11.5 inches wide, 6.125 inches tall, or .25 inches thick, but less than 12 by 15 by .75 inches.
- **Parcels** are pieces that are larger than 12 by 15 inches, or thicker than .75 inches.

Because of the difficulty involved in recording mail piece characteristics in the Household Diary, these categories do not correspond precisely to the shape categories used by Household Diary Study respondents.

Table 1.3 is derived from Table 1.2. It shows the revenue per piece and weight per piece for each subclass of mail by shape.

Table 1.1:
Total Mail Volume: FY 1987, 2003, 2004 and 2005
(Billions of Pieces)

Mail Classification	1987	2003	2004	2005
First-Class Mail:	**78.6**	**99.1**	**97.9**	**98.1**
First-Class Letters:	75.3	93.8	92.5	**92.4**
Single-Piece	53.9	46.7	45.2	**43.4**
Workshare	21.4	47.1	47.3	**49.1**
First-Class Cards:	3.2	5.2	5.4	**5.6**
Single-Piece	2.5	2.6	2.5	**2.5**
Workshare	.8	2.7	2.9	**3.1**
Priority Mail	**.4**	**.9**	**.8**	**.9**
Express Mail	**.0**	**.1**	**.1**	**.1**
Periodicals	**10.3**	**9.3**	**9.1**	**9.1**
Standard Mail:	**59.4**	**89.5**	**95.6**	**100.9**
Regular and ECR:	48.3	75.1	81.1	**85.9**
Regular	21.7	46.0	50.8	**53.9**
Enhanced Carrier Route	26.6	29.1	30.3	**32.0**
Nonprofit and NP ECR:	10.9	14.4	14.4	**15.0**
Nonprofit	8.6	11.4	11.8	**12.0**
NP Enhanced Carrier Route	2.3	3.0	2.7	**3.1**
Package Services:	**.6**	**1.1**	**1.1**	**1.2**
Parcel Post	.1	.4	.4	**.4**
Other Domestic Mail	**.5**	**.5**	**.6**	**.7**
International Mail	**.8**	**.9**	**.8**	**.9**
Total	**153.2**	**201.3**	**206.1**	**211.7**

Source: RPW Reports.
Note: Totals may not sum due to rounding.

Table 1.2:
Total Mail: Revenue, Pieces, and Weight by Shape, FY 2005

Mail Classification	Revenue *(Millions of Dollars)*				Pieces *(Millions of Pieces)*				Weight *(Millions of Pounds)*			
	Letters	**Flats**	**Parcels**	**Total**	**Letters**	**Flats**	**Parcels**	**Total**	**Letters**	**Flats**	**Parcels**	**Total**
First-Class Mail:	**30,703**	**4,374**	**705**	**35,782**	**93,094**	**4,482**	**495**	**98,071**	**3,321**	**889**	**160**	**4,370**
First-Class Letters:	29,529	4,374	705	34,608	87,465	4,482	495	92,442	3,270	889	160	4,319
Single-Piece	15,177	3,785	700	19,662	39,317	3,572	487	43,376	1,154	763	158	2,075
Workshare	14,352	588	6	14,946	48,148	910	8	49,066	2,116	127	1	2,243
First-Class Cards:	1,174	—	—	1,174	5,629	—	—	5,629	51	—	—	51
Single-Piece	599	—	—	599	2,521	—	—	2,521	17	—	—	17
Workshare	575	—	—	575	3,108	—	—	3,108	34	—	—	34
Priority Mail	**83**	**1,101**	**3,448**	**4,632**	**22**	**275**	**591**	**887**	**3**	**243**	**1,445**	**1,691**
Periodicals	**26**	**2,113**	**2**	**2,141**	**160**	**8,908**	**2**	**9,070**	**11**	**4,008**	**7**	**4,026**
Standard Mail:	**10,062**	**8,346**	**437**	**18,844**	**60,330**	**40,011**	**601**	**100,942**	**2,850**	**8,481**	**326**	**11,657**
Regular and ECR:	8,773	7,790	426	16,989	49,176	36,137	583	85,895	2,304	7,937	317	10,558
Regular	7,652	3,568	426	11,646	41,368	11,979	582	53,929	1,925	2,944	317	5,186
Enhanced Carrier Route	1,121	4,221	0	5,343	7,808	24,158	1	31,966	379	4,993	0	5,372
Nonprofit and NP ECR:	1,288	556	11	1,855	11,154	3,874	18	15,047	546	544	8	1,098
Nonprofit	1,188	372	11	1,571	9,921	2,050	18	11,990	495	351	8	854
NP Enhanced Carrier Route	100	184	0	284	1,233	1,824	0	3,057	50	193	0	244
Packages Services:	**—**	**293**	**1,903**	**2,196**	**—**	**303**	**863**	**1,166**	**—**	**404**	**3,256**	**3,660**
Parcel Post	—	11	1,220	1,231	—	3	385	388	—	4	1,912	1,916
Total	**40,873**	**16,226**	**6,496**	**63,595**	**153,606**	**53,979**	**2,551**	**210,136**	**6,184**	**14,024**	**5,194**	**25,403**
Other:												
Express Mail				872				55				52
Other Domestic Mail				1				700				145
International Mail				1,480				852				242
Total				**65,948**				**211,743**				**25,842**

Source: RPW Reports.
Note: Revenue data do not include fees. Totals may not sum due to rounding.

Table 1.3:
Total Mail: Revenue and Weight per Piece by Shape, FY 2005

Mail Classification	Revenue per Piece *(Dollars)*				Weight per Piece *(Ounces)*			
	Letters	Flats	Parcels	Total	Letters	Flats	Parcels	Total
First-Class Mail:	**.330**	**.976**	**1.425**	**.365**	**.571**	**3.174**	**5.157**	**.713**
First-Class Letters:	.338	.976	1.425	.374	.598	3.174	5.157	.748
Single-Piece	.386	1.060	1.437	.453	.470	3.416	5.204	.766
Workshare	.298	.647	.705	.305	.703	2.225	2.441	.732
First-Class Cards:	.209	—	—	.209	.145	—	—	.145
Single-Piece	.238	—	—	.238	.108	—	—	.108
Workshare	.185	—	—	.185	.175	—	—	.175
Priority Mail	**3.785**	**4.004**	**5.838**	**5.219**	**1.848**	**14.130**	**39.152**	**30.481**
Periodicals	**.163**	**.237**	**1.039**	**.236**	**1.067**	**7.198**	**63.801**	**7.101**
Standard Mail:	**.167**	**.209**	**.727**	**.187**	**.756**	**3.391**	**8.672**	**1.848**
Regular and ECR:	.178	.216	.731	.198	.750	3.514	8.713	1.967
Regular	.185	.298	.731	.216	.745	3.932	8.716	1.539
Enhanced Carrier Route	.144	.175	.477	.167	.777	3.307	5.677	2.689
Nonprofit and NP ECR:	.116	.144	.592	.123	.783	2.247	7.360	1.168
Nonprofit	.120	.181	.593	.131	.799	2.738	7.366	1.140
NP Enhanced Carrier Route	.081	.101	.362	.093	.654	1.696	6.401	1.276
Packages Services:		**.967**	**2.206**	**1.884**		**21.334**	**60.396**	**50.245**
Parcel Post		3.370	3.173	3.175		17.844	79.543	79.040
Total	**.266**	**.301**	**2.546**	**.303**	**.644**	**4.157**	**32.574**	**1.934**
Other:								
Express Mail				15.855				15.127
Other Domestic Mail				.001				3.318
International Mail				1.737				4.545
Total				**.311**				**1.953**

Source: RPW Reports.

Mail Flows

Mail volume can be broken into four basic flows, based on where it originates and destinates. These flows are:

- Household to household,
- Household to nonhousehold,
- Nonhousehold to household, and
- Nonhousehold to nonhousehold.

Table 1.4a shows the total mail in each flow, and Table 1.4b shows pieces per household per week.

Table 1.4a: Total Mail Flows
(Billions of Pieces)

Originating In:	Destinating In:		
	Household	Nonhousehold	Total Originating
Household	6.2	15.4	21.7
Nonhousehold	145.4	43.9	189.2
Total Destinating	**151.6**	**59.3**	**210.9**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Table 1.4b: Mail Flows per Household per Week

Originating In:	Destinating In:	
	Household	Nonhousehold
Household	1.1	2.6
Nonhousehold	24.7	N/A

Source: Household Diary Study, FY 2005.

Household Mail

As shown in Table 1.4a, mail to and from households constituted almost 80 percent of total mail volume in 2005. This equates to over 28.4 pieces per week sent and received by households in the U.S. Table 1.5 presents the volumes of mail sent and received by households as estimated from the Household Diary Study. The table shows the categories in which the households record their mail. Households received 151.6 billion pieces of mail and sent 21.7 billion. Both of these totals include the 6.2 billion pieces of mail that households sent to each other. The total mail received or sent by households in FY 2005 was 167.0 billion pieces.

In FY 2005, households received 15.6 billion pieces of unaddressed mail. These pieces were predominately the flyers and inserts associated with Standard Mail pieces, and are not included in the volume totals.

Table 1.5: Mail Sent and Received by Households
(Billions of Pieces)

Mail Classification	Received	Sent
First-Class Mail	58.8	21.1
Standard Regular Mail	70.1	—
Standard Nonprofit Mail	13.4	—
Periodicals	6.7	—
Packages	2.1	.3
Expedited	.5	.2
Total	**151.6**	**21.7**
Household to Household	6.2	
Total Mail Sent and Received by Households	167.0	
FY 2004 RPW Total	210.9	
Nonhousehold to Nonhousehold (Residual)	43.9	
Unaddressed	15.6	—

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Table 1.6 presents these data in two other forms, annual volumes per household and pieces per household per week. Many of the results presented subsequently in this report are presented in terms of pieces per household per week.

Table 1.6: Pieces Sent and Received per Household

Classification	Annual Pieces per Household	Pieces per HH per Week
Mail Received		
First-Class Mail	520	10.0
Standard Regular Mail	621	11.9
Standard Nonprofit Mail	119	2.3
Periodicals	59	1.1
Packages	19	.4
Expedited	4	.1
Total Mail Received	**1,342**	**25.8**
Mail Sent		
First-Class Mail:	187	3.6
Packages	3	.1
Expedited	2	.0
Total Mail Sent	**192**	**3.7**
Unaddressed	138	2.6

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Classes and Markets

- **First-Class Mail** is used to send transactional mail, correspondence and advertising. Because it is limited to pieces weighing thirteen ounces or less, it is primarily composed of letters and cards.
- **Standard Mail** is advertising mail. For the most part, Standard Mail is composed of letters and flats, although it contains a few postcards and packages as well.
- **Periodicals** are magazines and newspapers, and are predominantly flat shaped.
- **Priority Mail** and **Express Mail** are expedited services for delivering correspondence, transactional mail, and merchandise. Priority and Express pieces can be of any shape except postcards.
- **Package Services** is used to deliver merchandise, books, catalogs, and media such as CDs and DVDs. Most of this mail is parcel shaped, but there are a few flats as well.

Table 1.7 crosswalks between classes of mail and the markets they serve.

Table 1.7:
Mail Sent and Received by Households

Class	Market *(Billions of Pieces)*					
	Correspondence	Transactions	Advertising	Periodicals	Packages	Total
First Class	16.6	42.7	18.1	—	.7	70.6
Standard Mail	—	—	83.5	—	.7	84.2
Periodicals	—	—	—	6.7	—	6.7
Priority	—	—	—	—	.4	.4
Express	—	—	—	—	.1	.1
Package Services	—	—	—	—	.5	.5
Total	**16.6**	**42.7**	**101.6**	**6.7**	**2.4**	**162.5**

Source: Household Diary Study, FY 2005.
Note: Correspondence and Transactions include 7.5 billion pieces of secondary advertising mail also reported in Advertising Mail.
The " Total" column does not include pieces that could not be identified according to markets.
Package Volumes include ground packages and expedited.
Excludes Unclassified Mail.

Report Organization

The rest of the Household Diary Study report is organized around the markets the mail serves. Each chapter contains an analysis of the trends in the Household Diary Study data, as well as a discussion of how those trends impact and are impacted by changes in the broader market. The following provides an overview of each chapter.

Chapter 2: Profile of Mail Usage gives an analysis of household demographics. This chapter examines demographic trends over time and their impact on the mail, and discusses attributing factors, such as access to technology and changing attitudes.

Chapter 3: Correspondence examines mail that is used solely or primarily to deliver (non-sales-related) communications, such as letters and greeting cards. This chapter includes analysis of both personal and business correspondence.

Chapter 4: Transactions reviews financial transactions in the mail, and the impact that new technologies are having on that market. It analyzes household bill payment trends with a focus on technological and demographic change.

Chapter 5: Advertising Mail presents the trends in mail used to deliver sales-related messages. It contains information on household attitudes towards advertising by various media, treatment of advertising mail, and demographic determinants of advertising mail receipt.

Chapter 6: Periodicals examines magazines and newspapers delivered in the mail. It looks at how changing demographics are affecting the market for periodicals, and what the implications are for future volume.

Chapter 7: Packages analyzes household use of various types of parcels and express packages. It discusses the household market for merchandise delivery and looks at the market shares of the various parcel delivery companies.

In addition, there are three appendices to the report:

Appendix A contains a set of comparative tables for FY 1987, 2004, and 2005, organized by class of mail. A concordance is presented for comparison with pre-2000 reports.

Appendix B documents the study methodology and discusses how the data were collected, weighted and adjusted, and compares demographic data in the sample to that of the population as a whole.

Appendix C contains the instruments used to administer the survey.

Chapter 2: Profile of Mail Usage

Introduction

This chapter provides information on demographic trends and other factors affecting mail volume, providing a basis for assessing mail volume growth. The breakouts developed in this chapter provide the basis for much of the analyses in subsequent chapters.

The first section looks at growth in mail volume, population, households, and delivery points over recent decades. The next section examines the demographic characteristics of mail users, contrasting higher mail volume households with lower volume households. The third section details the emerging demographic trends that will affect the future of mail. The last section examines some of the technological, behavioral, and attitudinal factors affecting mail.

Mail Volume and Demographics

Total U.S. mail volume grew from 110 billion pieces in GFY 1981 to 212 billion in 2005, an increase of 93 percent. This growth has outpaced the rate of population growth and household formation. Over the same period, adult population grew 39 percent and households grew by 34 percent. The number of places to which the Postal Service delivers to increased still faster, growing by 47 percent.

As Table 2.1 shows, however, volume growth has slowed dramatically over the last five years, while growth in population and households was much steadier.

Total U.S. mail volume grew from 110 billion pieces in GFY 1981 to 212 billion in 2005, an increase of 93 percent, outpacing population growth and household formation.

The 1980s was a time of extraordinary mail volume growth that started in 1978 and continued through 1988. In 1984, mail volume grew more than ten percent. During this period, technology facilitated this growth. Construction of computerized databases and techniques for sorting large amounts of data created a fertile climate for direct mail marketing. Computerization of financial systems encouraged billing by mail and payments through the mail. These innovations in business processes were further encouraged by postal rates. The Postal Service introduced worksharing discounts, encouraging mailers to prepare the mail in ways that reduced the total system cost of creating and delivering the mail. Mailers could take advantage of these discounts by sorting the mail in advance. The Postal Service would get the mail presorted to the individual ZIP codes and/or to the carrier routes associated with those ZIP codes.

In the late 1980s and early 1990s, mail volume growth barely kept pace with household growth. The demand for mail was hurt by recession and two very large rate increases. This was also a period in which the Postal Service absorbed substantial costs that were reapportioned from the Federal government's retirement programs.

Table 2.1:
Mail Volume and Demographics
Average Annual Growth 1981-2005

	1981-1990	1991-2000	2001-2005
Total Mail Volume	4.6%	2.3%	.5%
Delivery Points	1.7%	1.5%	1.2%
Adult Population	1.5%	1.3%	1.2%
Households	1.4%	0.9%	1.1%

Source: U.S. Postal Service, U.S. Census Bureau.

The last half of the 1990s saw rapid growth in mail volume, spurred by a strong economy and rates that increased by less than inflation. The Postal Service also realigned the incentives built into its price structure. It reduced the incentives mailers had for presorting mail and encouraged them to prebarcode it. By 2002, the majority of letters the Postal Service received had qualifying barcodes on them. This restructuring of the rates took advantage of the extensive automation of mail preparation and sorting that occurred over the previous decade.

The 1990s ended in a speculative bubble as the U.S. economy rapidly embraced information technology and integrated the Internet into its business processes. An economic recession followed that began in March 2001. The 2001 Government Fiscal Year ended with the terrorist attacks on the World Trade Center and the Pentagon. This led to large-scale disruptions of those mail services dependent on air transport such as First-Class, Priority, and

Express Mail. When air service was restored, Priority Mail was no longer allowed on commercial passenger flights. GFY 2002 began with bio-terrorism. Lethal anthrax sent through the mail resulted in five deaths and a number of serious injuries. The 2.2 percent mail volume decline in 2002 was the worst since World War II. In 2003, Standard Mail volume recovered to a new high but total First-Class volume continued to decline. Workshared First-Class Mail fell for the first time ever. Since 2003, Standard Mail volume grew along with the economy, reaching new highs and exceeding First- Class Mail for the first time in 2005. Total First-Class volume, on the other hand, has continued to decline in part due to the diversion of bills and statements to electronic alternatives and to lower cost Standard Mail options.

For the period 2001-2005, total mail volume grew a modest 2.1 percent while First-Class volume fell 5.4 percent. Nevertheless, population and households grew approximately 4 percent during the same period. The Postal Service added 4.8 percent more delivery points to its network.

Continued growth in delivery points will be an ongoing source of pressure on Postal costs.

The Postal Service depends on mail volume growth to fund universal service. The number of addresses the delivery network serves increases as the number of American businesses and households grow. When mail volume grows faster than the number of delivery points, the system benefits from significant economies of scale. When mail volume grows slower, the Postal Service's ability to fund delivery service is hampered because the Postal Service charges its customers for piece volume but does not assess connect charges, access fees, or system fees, like many other network enterprises.

Over the next five years, both the total U.S. population and the adult population are projected to grow just under one percent a year. This is in line with the recent past. Households are expected to grow slightly faster: about 1.2 percent a year. This should result in continued growth in the number of addresses the Postal Service serves, and in the resulting costs. Given recent mail volume trends, this presents a challenge. Continued growth in delivery points that exceeds volume growth could be an ongoing source of pressure on postal costs.

Characteristics of Higher- and Lower-Volume Households

Tables 2.2 and 2.3 show the demographic characteristics of households by the amount of mail they receive. It is immediately obvious from these tables that household mail use is strongly correlated with both income and education. Note, however, the similar correlation between mail receipt and Internet access, also related to income and education. Therefore, households that make the most use of the mail are the households with the greatest opportunity to use alternatives to the mail.

These high-volume households are beginning to take advantage of the opportunity to move away from the mail. Households that receive 30 or more pieces of mail each week pay 14.6 percent of their bills electronically, up from 7.9 percent in 2003. In comparison, households that receive less than 30 pieces of mail each week paid 9.5 percent of their bills electronically, up from 5.3 percent in 2003.

Table 2.2:
Characteristics of Higher- and Lower-Mail-Volume Households

Mail Received *(Pieces per HH per week)*	Households *(Millions)*	Median Annual HH Income	HHs w/ Internet Access *(Percent)*	Annual Bills Paid *(Millions)*	Annual Bills Paid Electronically *(Millions)*	Mail Sent *(Pieces per HH per week)*
45 or more	20.0	$78,409	89%	3,672	619	5.3
36-44	15.2	$64,720	81%	2,424	288	4.0
30-35	14.7	$60,566	78%	2,196	303	3.0
24-29	17.4	$52,234	74%	2,460	250	3.0
18-23	19.3	$45,901	68%	2,555	256	2.4
12-17	14.1	$30,110	60%	1,663	135	2.0
Less than 12	12.4	$25,036	46%	1,247	114	1.3
Total	**112.9**	**$52,858**	**72%**	**16,217**	**1,965**	**3.1**

Source: Household Diary Study, FY 2005.
Note: Annual mail includes USPS and Non-USPS mail.

Table 2.3:
Education of Higher- and Lower-Mail-Volume Households

Mail Received *(Pieces per HH per week)*	Households *(Millions)*	Educational Attainment of Head of Household			
		Less than High School	High School Graduate	Some College or Technical School	College Graduate
45 or more	20.0	7%	23%	24%	46%
36-44	15.2	9%	29%	23%	38%
30-35	14.7	11%	29%	27%	33%
24-29	17.4	12%	33%	31%	24%
18-23	19.3	17%	37%	27%	19%
12-17	14.1	21%	33%	29%	17%
Less than 12	12.4	33%	30%	25%	12%
Total	**112.9**	**15%**	**31%**	**27%**	**28%**

Source: Household Diary Study, FY 2005.
Note: Percentages do not total 100% due to heads of households who did not answer the educational attainment question. Mail received includes USPS and Non-USPS mail. Percentages in this table are row percentages.

Demographic Characteristics of U.S. Households

This section develops breakouts of households by demographic categories that influence the volume of mail sent and received. It looks at both traditional and newly emerging factors. The following chapters will show how mail volume varies with these household characteristics.

Income, Education, and Age

Traditionally, mail use was largely determined by household income, education and age. As Table 2.4 shows, income and education are strongly correlated with each other, as expected.

The relationship between income and age, shown in Table 2.5, is somewhat more complicated. Up to retirement, household income and age are fairly closely related. After retirement, households earn substantially less, although by that point mail behavior is pretty well set, and older households continue to receive similar amounts of advertising and periodicals, and pay similar amounts of bills, even though their income declines.

Table 2.4:
Households by Income and Education
(Millions of Households)

Household Income *(Thousands)*	Educational Attainment of Head of Household					Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	Don't Know/ Refused	
Under $35	9.4	11.5	8.4	3.5	.3	33.0
$35 to $65	4.0	10.9	10.0	7.5	.4	32.7
$65 to $100	1.6	6.7	6.4	9.5	.0	24.2
Over $100	.2	2.0	3.0	7.5	—	12.8
Don't know / Refused	1.3	3.1	2.2	3.3	.3	10.2
Total	**16.6**	**34.2**	**29.9**	**31.2**	**1.0**	**112.9**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Table 2.5:
Households by Income and Age
(Millions of Households)

Household Income *(Thousands)*	Age of Head of Household				Total
	Under 34	35 to 54	Over 55	Don't Know/ Refused	
Under $35	9.0	8.5	15.5	.0	33.0
$35 to $65	8.6	13.4	10.7	.1	32.7
$65 to $100	5.4	13.0	5.8	.0	24.2
Over $100	1.8	8.0	2.9	.0	12.8
Don't know / Refused	1.0	3.4	5.3	.5	10.2
Total	**25.7**	**46.3**	**40.2**	**.7**	**112.9**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Household Size

The majority of U.S. households include either one or two adults, but households with three or more adults make up almost 17 percent of the total. Nuclear families, once considered the norm, account for only about 22 percent of households. The changing composition of households has impacted the amount and kinds of mail sent and received by households over the past 20 years, generating more and different kinds of advertising mail, as well as affecting transaction mail trends (bills tend to be tied to households as much as to individuals).

Table 2.6:
Households by Size
(Millions of Households)

Household Size	
One person	18.3
Two	41.6
Three	21.8
Four	18.2
Five or more	12.8
Don't know / Refused	.2
Total	**112.9**

Source: Household Diary Study, FY 2005.
Note: Total may not sum due to rounding.

Table 2.7:
Households by Number of Adults
(Millions of Households)

Number of Adults	
One	22.1
Two	72.0
Three or more	18.9
Total	**112.9**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

Table 2.8:
Households by Type of Internet Access
(Millions of Households)

Type of Internet Access	
Broadband	36.6
Dial-up	42.6
None	31.4
Don't Know / Refused	2.3
Total	**112.9**

Source: Household Diary Study, FY 2005.
Note: Totals may not sum due to rounding.

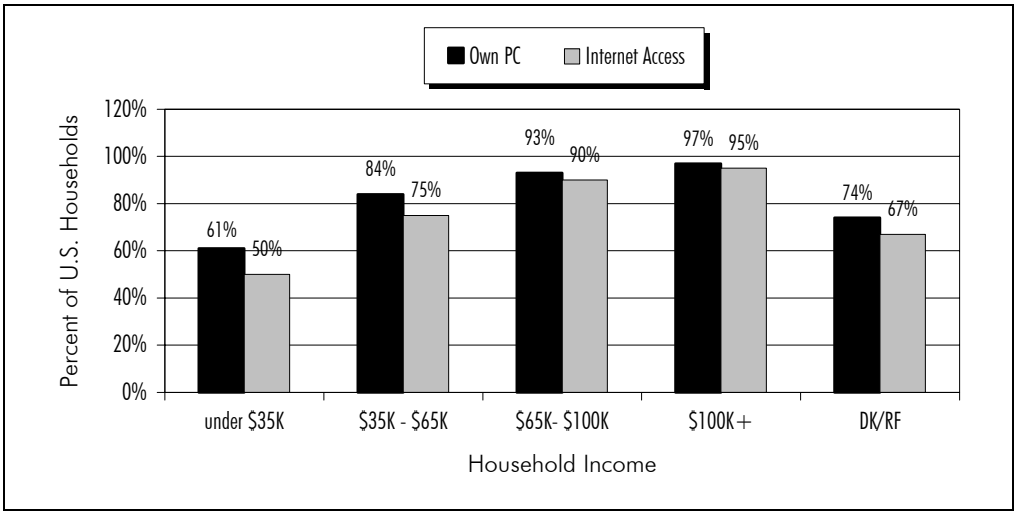
Internet Access

Access to and use of new technologies such as personal computers and the Internet have a large and growing impact on mail use. Although a significant number of pieces both sent and received by households are still related to bills and statements, electronic activity in this area is diverting mail once used for these purposes. On the other hand, on-line shopping potentially adds packages and catalog delivery to the Postal Service mailstream.

Table 2.8 shows that almost 80 percent of households have Internet access. The highest levels of access are within households with incomes over $100,000 in which 95 percent have Internet access, as can be seen in Figure 2.1. Fifty percent of households with incomes under $35,000 have Internet access, up from 41 percent in 2004. For households without Internet access, 45 percent indicated they planned to purchase the service within the next year.

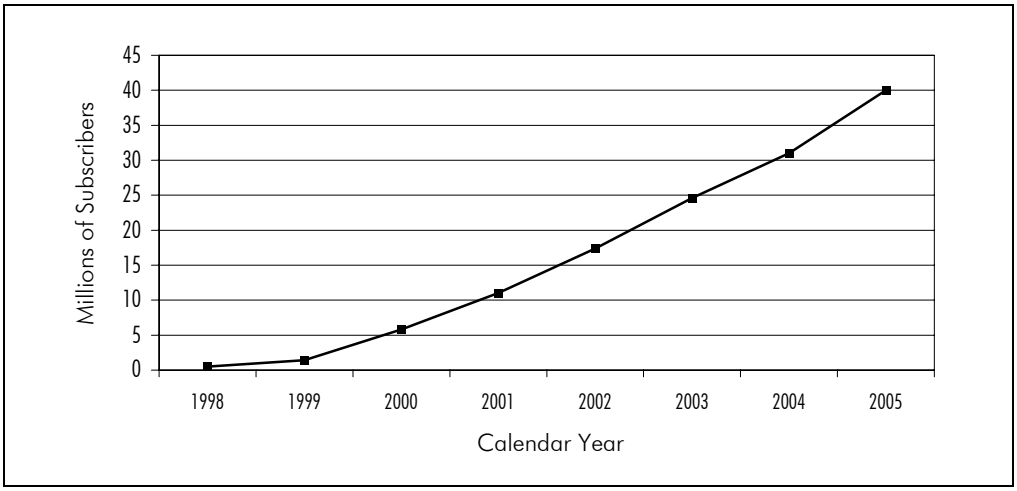
Forty-eight percent of wired households, or 32 percent of the total, have broadband access (DSL, cable, or T1). Figure 2.2 shows the trend in broadband connections. The rapid growth of broadband expands the potential scope of electronic diversion of the mail. A fast, always-on connection to the Internet becomes a stronger alternative channel for the delivery of entertainment, information, and communication. As more households begin using broadband, we expect to see effects not only on bill payments, but also bill and statement presentment, periodicals, and perhaps even advertising mail.

Figure 2.1:
PC Ownership and Internet Access



Source: Household Diary Study, FY 2005.

Figure 2.2:
Broadband Subscribers



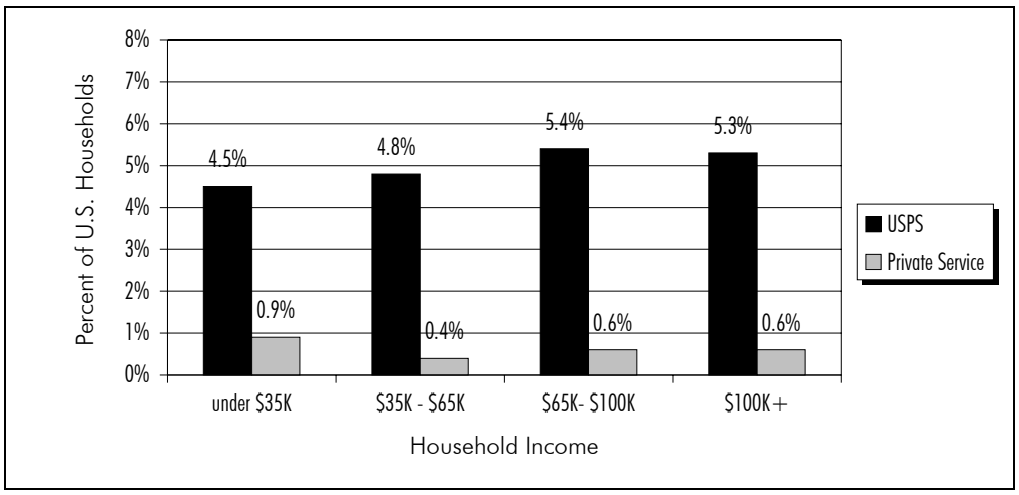
Source: Leichtman Research Group.

Use of the Post Office

A rented mailbox is one alternative that households use to manage their mail. Households that rent mailboxes overwhelmingly use the Postal Service for this service, with less than one percent of all households in the U.S. renting a box from a private company. As seen in Figure 2.3, higher income households are slightly more likely to use a post office box than lower income households, a pattern similar to prior years. Overall post office box use, however, declined over the past three years, after 2001, largely as a result of price increases, with less than five percent of U.S. households renting a box from the Postal Service in 2005, compared to ten percent in 2001.

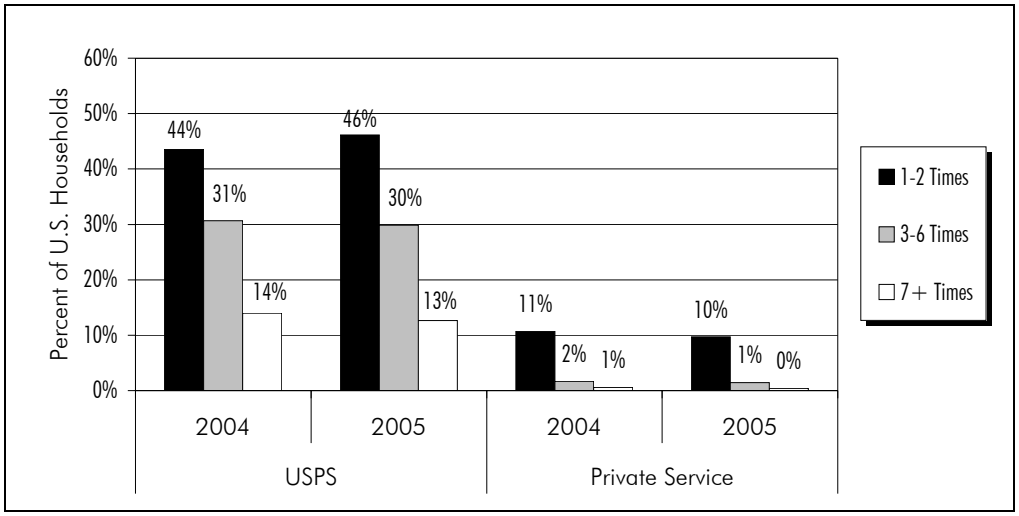
The Postal Service currently owns and operates over 37 thousand post office locations throughout the United States. As shown in Figure 2.4, the use of post offices for mailing services continues to dominate the mail service industry. Over 89 percent of all U.S. households patronize a post office at least once a month, while just eleven percent visit a private mailing company. Over 43 percent of all households in the U.S. visit the post office three or more times in a month. Even with the continued availability of mail-related products and services through alternative modes (such as Internet orders), in-person visits to postal facilities remain stable. However, visits to private mailing companies declined from 14 percent in 2004.

Figure 2.3:
Household Use of Rented Mailboxes



Source: Household Diary Study, FY 2005.

Figure 2.4:
Household Visits to Post Office in Past Month



Source: Household Diary Study, FY 2004 and 2005.
Note: Years sum to less than 100% due to exclusion of zero responses.

Chapter 3: Correspondence

Introduction

This chapter examines correspondence mail among households and between households and businesses including letters, greeting cards, invitations, and announcements. In several cases, this chapter and several following it examine comparisons in data between 2003 and 2005, providing a better illustration of mail trends over time

Correspondence Mail Volume

Total correspondence sent and received represents about 10 percent of all household mail volumes (see Table E-2). Personal correspondence mail has declined for more than 15 years. In 1987, households reported receiving about 2.3 pieces of correspondence mail per week, of which 1.6 pieces were personal correspondence. By 2005, personal correspondence received declined by about a third, to 1.0 piece per household per week, while overall correspondence received increased by about 0.2 pieces per week. Table 3.1 provides a recent history of total correspondence volumes.

To a large extent, the declines in correspondence mail were driven by changes in communication technology over that period. First, affordable long distance telephone service provided an alternative to personal letters and business inquiries, and then e-mail and the Internet completely transformed the marketplace.

Correspondence Mail and Household Characteristics

The following tables break down correspondence mail sent and received by households using the demographic categories developed in Chapter 2.

Income, Education and Age

Table 3.2 shows that both household income and educational attainment have a strong effect on correspondence received. However, as Table 3.3 shows, within a given educational category, income has much less of an effect on the amount of correspondence mail a household sends.

Table 3.1:
First Class Correspondence Mail Sent and Received by Sector

Sector	Volume *(Millions of Pieces)*			Change, 2003-2005
	2003	2004	2005	
HH to HH	6,457	6,561	5,870	-9.1%
NHH to HH	8,414	8,446	8,647	2.8%
HH to NHH	2,050	2,167	2,119	3.4%
Total	**16,921**	**17,173**	**16,639**	**-1.7%**
Sector	**Pieces per Household per Week**			**Share of 2005 Total**
	2003	**2004**	**2005**	
HH to HH	1.1	1.1	1.0	35%
NHH to HH	1.5	1.5	1.5	52%
HH to NHH	.4	.4	.4	13%
Total	**2.9**	**3.0**	**2.8**	

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Totals may not sum due to rounding.

Table 3.2:
Correspondence Mail Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	1.4	1.9	1.5	2.4	**1.7**
$35 to $65	1.9	2.4	2.5	2.6	**2.4**
$65 to $100	3.4	2.4	3.2	3.1	**2.9**
Over $100	.4	2.6	3.1	3.8	**3.4**
Total	**1.7**	**2.2**	**2.4**	**3.1**	**2.5**

Source: Household Diary Study, FY 2005.

Table 3.3:
Correspondence Mail Sent by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	.7	1.1	1.2	1.6	**1.0**
$35 to $65	.7	1.5	1.4	1.6	**1.4**
$65 to $100	1.5	1.1	1.9	1.5	**1.5**
Over $100	.6	1.4	1.4	1.8	**1.6**
Total	**.8**	**1.2**	**1.4**	**1.6**	**1.3**

Source: Household Diary Study, FY 2005.

Tables 3.4 and 3.5 show that age also has a considerable effect on correspondence mail sent and received by households. Regardless of their income, younger households both send and receive fewer pieces of correspondence mail. Young adults have traditionally sent and received less mail than older adults, but the advent of the Internet age widened the gap between these two age groups.

Table 3.4:
Correspondence Mail Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 34	35 to 54	Over 55	
Under $35	1.4	1.6	1.9	**1.7**
$35 to $65	1.9	2.6	2.7	**2.4**
$65 to $100	2.4	3.0	3.4	**2.9**
Over $100	2.3	3.3	4.4	**3.4**
Total	**1.9**	**2.6**	**2.6**	**2.5**

Source: Household Diary Study, FY 2005.

Table 3.5:
Correspondence Mail Sent by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 34	35 to 54	Over 55	
Under $35	.8	.9	1.2	**1.0**
$35 to $65	.9	1.4	1.6	**1.4**
$65 to $100	1.6	1.3	2.0	**1.5**
Over $100	1.1	1.5	2.4	**1.6**
Total	**1.0**	**1.3**	**1.6**	**1.3**

Source: Household Diary Study, FY 2005.

Household Size

As expected, household size has a positive impact on correspondence mail. Tables 3.6 and 3.7 show that the jump from one person to two is associated with considerable jumps in correspondence mail volume, although further increases in size do not have nearly the same effect. Tables 3.8 and 3.9 show that these increases are generally due to the presence of an additional adult in the household.

Table 3.6:
Correspondence Mail Received by Household Size
(Pieces per Household per Week)

Household Size	
One person	1.6
Two	2.5
Three	2.4
Four	3.0
Five or more	3.1
Total	**2.5**

Source: Household Diary Study, FY 2005.

Table 3.7:
Correspondence Mail Sent by Household Size
(Pieces per Household per Week)

Household Size	
One person	1.1
Two	1.5
Three	1.2
Four	1.3
Five or more	1.7
Total	**1.4**

Source: Household Diary Study, FY 2005.

Table 3.8:
Correspondence Mail Received by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults	
One	1.6
Two	2.6
Three or more	3.0
Total	**2.5**

Source: Household Diary Study, FY 2005.

Table 3.9:
Correspondence Mail Sent by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults	
One	1.1
Two	1.4
Three or more	1.5
Total	**1.4**

Source: Household Diary Study, FY 2005.

Internet Access

Households with Internet access also tend to send and receive more correspondence mail than households without. The explanation for this somewhat counterintuitive result is the high correlation among income, educational attainment, and the presence of an Internet connection in the home. Households with Internet access have a greater average income ($75,000 for Broadband and $58,000 for Dial-up users) than households without ($36,000). Similarly, households with Internet access have a higher level of education (42% of Broadband users and 28% of Dial-up users have a

college degree) than households without (11% have a college degree). In fact, these correlations could be a warning sign for mail, since more volume goes to households that are vulnerable to diversion. Table 3.11 shows that, although the number of mail pieces sent by households with Broadband is higher than for household without access, it is also lower than the number for Dial-up users. This may indicate that diversion increases as households upgrade to faster technologies.

Table 3.10:
Correspondence Mail Received by Type of Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	2.7
Dial-up	2.7
None	1.7
Total	**2.5**

Source: Household Diary Study, FY 2005.

Table 3.11:
Correspondence Mail Sent by Type of Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	1.3
Dial-up	1.5
None	1.0
Total	**1.3**

Source: Household Diary Study, FY 2005.

Personal Correspondence

In FY 2005, personal correspondence accounted for an average 1.0 pieces per week, .1 less then in 2004. Table 3.12 shows the total volumes and average number of pieces by personal correspondence type.

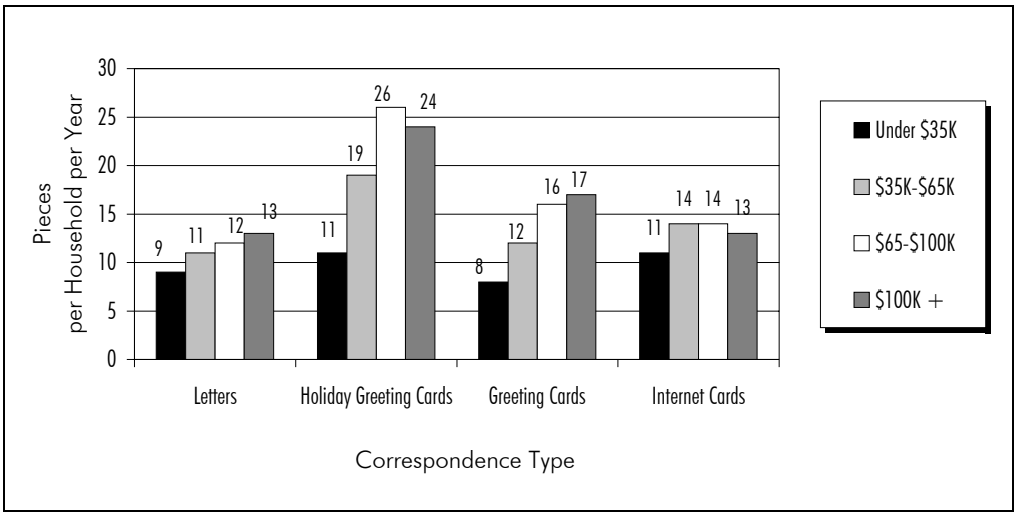
The drop in personal letters accounts for most of the per-household decline in personal correspondence mail between 2003 and 2005, continuing a long-term decline. The number of Internet greeting cards declined by about the same percentage as mailed correspondence.

Table 3.12: Personal Correspondence Sent and Received

Correspondence Type	Volume *(Millions of Pieces)*			Change, 2003-2005
	2003	2004	2005	
Personal Letters	1,468	1,385	1,227	-16.4%
Holiday Greeting Cards	2,196	2,417	2,169	-1.2%
Non-Holiday Greeting Cards	1,620	1,597	1,418	-12.5%
Invitations	665	728	657	1.2%
Announcements	183	136	124	-32.3%
Other Personal	326	298	275	-15.5%
Total	**6,458**	**6,561**	**5,870**	**-9.1%**
Internet Cards	1,541	1, 654	1, 399	-9.2%
Correspondence Type	**Pieces per Household per Week**			**Share of 2005 Total**
	2003	**2004**	**2005**	
Personal Letters	.3	.2	.2	17%
Holiday Greeting Cards	.4	.4	.4	30%
Non-Holiday Greeting Cards	.3	.3	.2	20%
Invitations	.1	.1	.1	9%
Announcements	.0	.0	.0	2%
Other Personal	.1	.1	.0	4%
Total	**1.1**	**1.1**	**1.0**	**81%**
Internet Cards	.3	.3	.2	19%

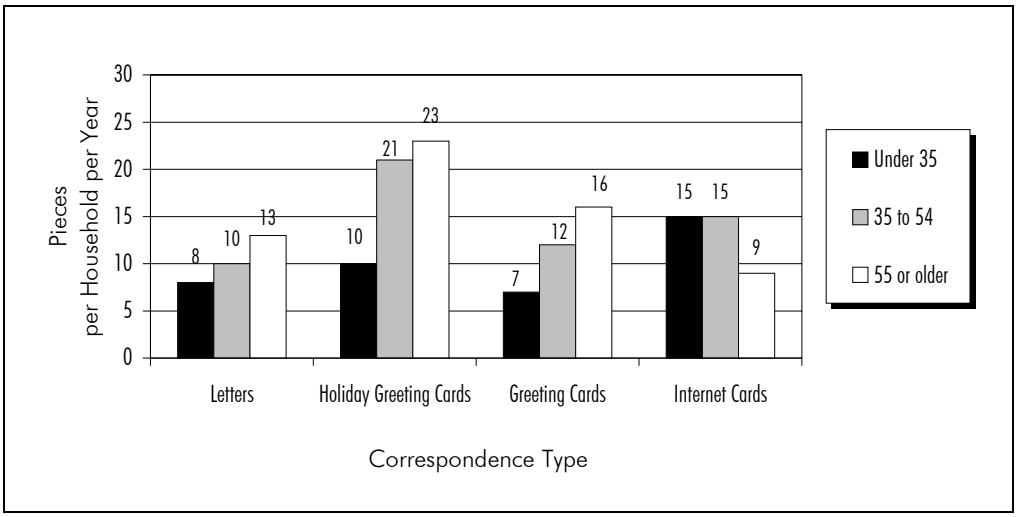
Source: Household Diary Study, FY 2003, 2004 and 2005. Note: Totals may not sum due to rounding.

Figure 3.1:
Weekly Personal Correspondence Sent by Income Group
(Pieces per Household per Year)



Source: Household Diary Study, FY 2005.

Figure 3.2:
Weekly Correspondence Sent by Age Cohort
(Pieces per Household per Year)



Source: Household Diary Study, FY 2005.

Figure 3.1 shows the major personal correspondence types by income. Personal correspondence sent by households varies with household income (Fig. 3.1). Households with incomes over $100,000 were more likely to send letters and holiday greeting cards than lower-income households in 2005. The largest disparity between high- and low-income households is in the volume of holiday greeting cards sent. Households with incomes over $100,000 sent an average of 24 holiday greeting cards during the 2004 holiday season (FY 2005), compared to the 11 sent by households with incomes under $35,000. There does not appear to be a strong relationship between household income and use of Internet greeting cards.

The number of letters and greeting cards sent varies across age groups (Fig. 3.2). Households where the head is 55 years of age or older sent more personal correspondence of all types through the mail. Households aged 55 or older sent an average of thirteen letters and 23 holiday greeting cards in 2005. They also sent an average of 16 non-holiday cards.

In comparison, households under the age of 35 sent an average of eight letters and ten holiday greeting

cards per year in FY 2005, along with seven non-holiday cards. Households whose heads are under 35 years old sent about fifty percent more (15) Internet cards than traditional, mailed non-holiday greeting cards in FY 2005, while households whose heads are 55 or over sent only half as many Internet cards as they sent traditional cards through the mail (9).

The relationship between holiday greeting cards received and income and age is shown in Figure 3.3. For most income category, the number of holiday greeting cards received in 2005 was down or in some cases the same as 2003. Households with incomes between $65,000 and $100,000 received the greatest number of holiday greeting cards during FY 2005 (25.5), up from 20.3 in FY 2003. The decrease from 2003 is most pronounced for households with incomes over $100,000.

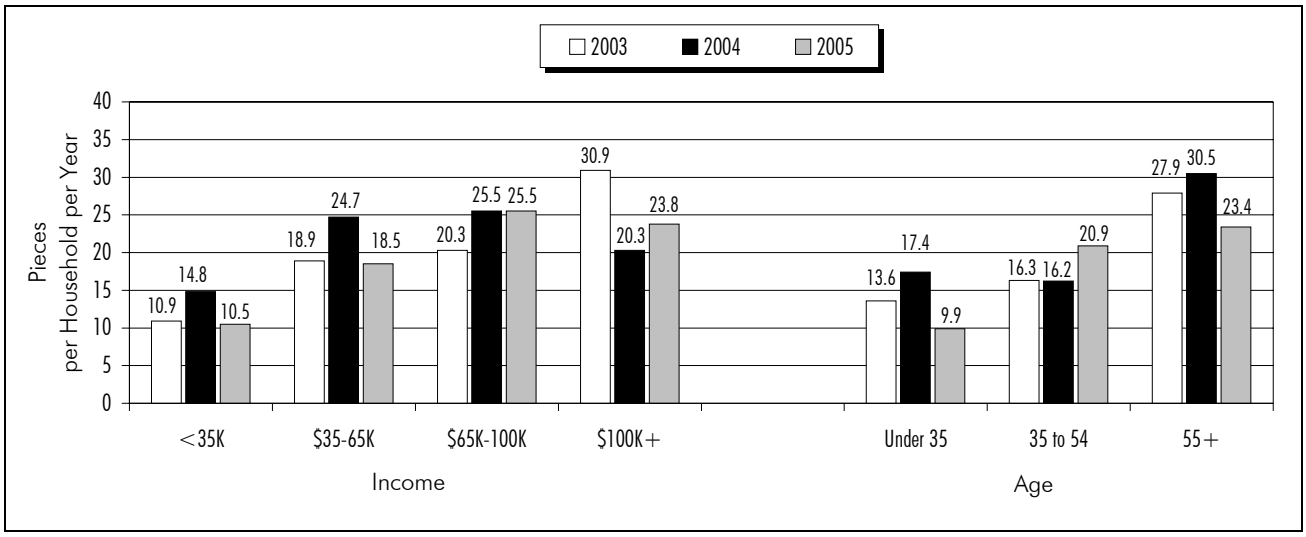
The number of holiday greeting cards received also decreased since 2003 for all age categories, with the exception of the 35 to 54 age group.

The largest average number of holiday greeting cards are received by households where the head of household is age 55 or older. These households received an average of 23.4 pieces in 2005 as compared to the 27.9 pieces received in 2003.

As shown in Table 3.13, households with Internet access (including both broadband and dial-up) receive more letters and greeting cards, compared to households without Internet access. These households also send more correspondence. As stated above, this is largely due to the correlation between income, education, and the presence of an Internet connection in the home.

In examining personal Internet use through the HDS Personal Technology Questionnaire, we find that 60.1 percent of all persons who have Internet access send at least one personal e-mail during a typical day. Figure 3.4 shows the number of personal e-mails sent and received in a typical day by those that have used e-mail in the past 12 months.

Figure 3.3:
Holiday Greetings Received by Age and Income, FY 2003, 2004 and 2005



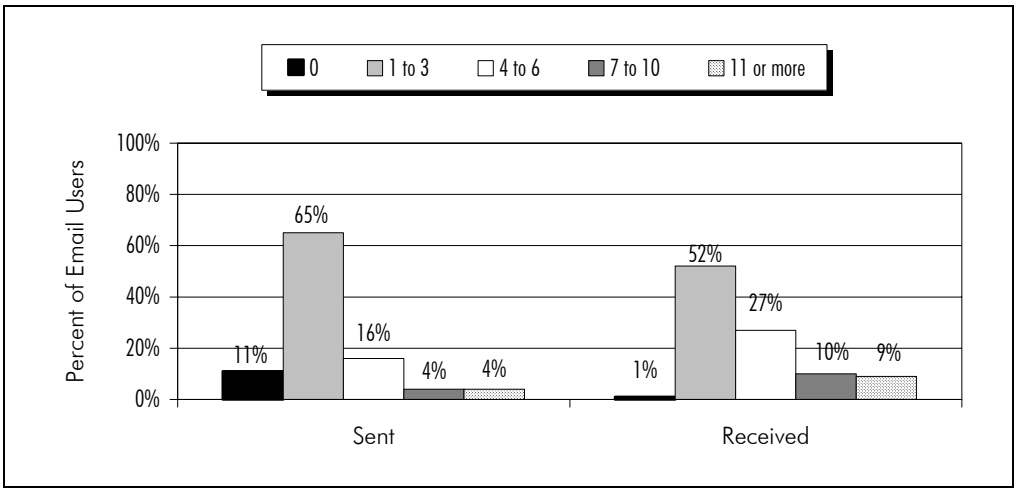
Source: Household Diary Study, FY 2003, 2004 and 2005.

Table 3.13:
Personal Correspondence by Type of Internet Access
(Pieces per Household per Week)

Correspondence Type	No Internet Access	Dial-up	Broadband
Personal Letters	.16	.23	.23
Holiday Greeting Cards	.29	.46	.32
Non-Holiday Greeting Cards	.18	.24	.29
Total	**.64**	**.93**	**.83**

Source: Household Diary Study, FY 2005.

Figure 3.4:
Daily Personal E-mails Sent and Received



Source: Household Diary Study, FY 2005.
Base: Persons 13 years or older in the United States that have communicated via e-mail during FY 2005.

Business Correspondence

Outside of bill payment, statements and advertising, which are discussed in Chapters 4 and 5, households and businesses exchange many types of correspondence. Table 3.14 outlines volumes by correspondence type for the years 2003 through 2005. Correspondence received from the non-household sector accounts for 52 percent of total correspondence volumes. Announcements and invitations comprise the largest correspondence volumes from nonhouseholds (6.0 billion).

Table 3.14: Correspondence Type by Sector
(Millions of Pieces)

Correspondence Type	2003	2004	2005	Change, 2003-2005
Business/Government/Social Received by Households				
Invitation/Announcement	3,995	4,657	4,161	4.2%
Holiday Greeting from Business	300	335	500	66.7%
Other Business/Government	1,214	1,132	1,667	37.3%
Total Business Received	**5,509**	**6,123**	**6,328**	**14.9%**
Announcement	2,629	2,042	1,827	-30.5%
Other Social	276	281	491	78.0%
Total Social Received	**2,905**	**2,323**	**2,318**	**-20.2%**
Total Received	**8,414**	**8,446**	**8,647**	**2.8%**
Business/Government/Social Sent by households				
Inquiry	664	626	549	-17.3%
Other Business/Government	946	1,094	1,153	21.9%
Total Business Sent	**1,610**	**1,720**	**1,702**	**5.7%**
Letter	110	193	178	61.7%
Inquiry	100	86	60	-39.9%
Other Social	231	168	179	-22.5%
Total Social Sent (Social includes social, political and nonprofit.)	**440**	**447**	**417**	**-5.2%**
Total Sent	**2,050**	**2,167**	**2,119**	**3.4%**

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Totals may not sum due to rounding.

Chapter 4: Transactions

Introduction

This chapter examines the volumes and trends in transactions mail; the bills, statements, payments, donations, rebates, and orders sent and received by households. Information is presented on household bill payment trends, especially in light of electronic alternatives to traditional mail payments.

Transactions Mail Volume

Transactions sent and received comprise about 25 percent of all household mail volumes (as seen in Table E-2) and 53 percent of household First-Class Mail; as such, they are an important part of the mailstream. Although many businesses now use electronic funds transfer (EFT) or other electronic technologies to settle transactions, households still pay a majority of their recurring bills through the mail. As the Internet becomes more ubiquitous, however, the movement towards consumer electronic bill presentment and payment (EBPP) is picking up.

Transactions comprise 53 percent of Household First-Class Mail

As shown in Table 4.1, overall transaction volumes have stabilized since 2003. The number of bills increased slightly in 2005, however, the number of charitable requests decreased somewhat, in effect making the change in mail volume from 2003 insignificant.

Table 4.1:
Transactions Mail Sent and Received

Transaction Type	Volume *(Millions of Pieces)*			Change, 2003-2005
	2003	2004	2005	
Business				
Bills	18,542	18,481	18,656	.6%
Bill Payments	10,707	11,152	10,809	1.0%
Statements	6,429	6,452	6,594	2.6%
Confirmations	2,429	2,252	2,518	3.7%
Payments (to HH)	1,618	1,552	1,437	-11.2%
Orders	739	734	769	4.1%
Rebates	185	207	206	11.3%
Total Business	**40,649**	**40,828**	**40,990**	**.8%**
Social/Charitable				
Requests for Donation	729	634	636	-12.7%
Donations	536	598	560	4.5%
Bill	465	182	184	-60.4%
Confirmations	319	306	324	1.5%
Total Social/Charitable	**2,049**	**1,721**	**1,704**	**-16.8%**
Total Transactions	**42,698**	**42,549**	**42,694**	**0.0%**

Table 4.1:
Transactions Mail Sent and Received (cont.)

Transaction Type	Pieces per Household per Week			Share of 2005 Total
	2003	2004	2005	
Business				
Bills	3.2	3.2	3.2	44%
Bill Payments	1.8	1.9	1.8	25%
Statements	1.1	1.1	1.1	15%
Confirmations	.4	.4	.4	6%
Payments (to HH)	.3	.3	.2	3%
Orders	.1	.1	.1	2%
Rebates	.0	.0	.0	0%
Total Business	**7.0**	**7.0**	**7.0**	**96%**
Social/Charitable				
Requests for Donation	.1	.1	.1	1%
Donations	.1	.1	.1	1%
Bill	.1	.0	.0	0%
Confirmations	.1	.1	.1	1%
Total Social/Charitable	**.4**	**.3**	**.3**	**4%**
Total Transactions	**7.4**	**7.4**	**7.3**	**100%**

Source: Household Diary Study, FY 2003, 2004 and 2005.

Transactions Mail and Household Characteristics

The following tables break down transactions mail sent and received by households based on the demographic categories introduced in Chapter 2.

Income, Education, and Age

As seen in Tables 4.2 and 4.3, household income and educational attainment influence the amount of transaction mail sent and received. The basis for this relationship is fairly clear; higher income and better-educated households, on average, have more financial accounts, insurance policies, and credit cards – all generators of transactions mail volume.

Table 4.2:
Transactions Mail Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	3.5	3.8	3.6	4.2	**3.7**
$35 to $65	5.2	5.2	5.1	5.2	**5.2**
$65 to $100	7.2	6.2	6.2	6.1	**6.1**
Over $100	3.3	6.4	7.5	7.5	**7.2**
Total	**4.4**	**5.0**	**5.1**	**6.0**	**5.2**

Source: Household Diary Study; FY 2005.

Table 4.3:
Transactions Mail Sent by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	1.2	1.6	1.7	1.9	**1.6**
$35 to $65	1.7	2.2	2.2	2.2	**2.1**
$65 to $100	3.2	2.8	2.2	2.4	**2.5**
Over $100	1.4	2.2	2.9	2.4	**2.4**
Total	**1.6**	**2.1**	**2.1**	**2.3**	**2.1**

Source: Household Diary Study, FY 2005.

Tables 4.4 and 4.5 also show that household age has a strong effect on transactions mail, independent of income. Across all income categories, households whose heads are under 35 years old send and receive less transactions mail. In part, this is due to the fact that younger households are less likely to own their home and have fewer insurance policies. However, it is also the case that these households are more active users of electronic alternatives to traditionally mail-based transactions. This is especially conspicuous for transactions mail sent by households (primarily bill payments), where households under 35 years old sent only about 65 percent as much mail as older households.

Across all income categories, households whose heads are under 35 years old send and receive less transactions mail.

Table 4.4:
Transactions Mail Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 35	35 to 54	Over 55	
Under $35	3.3	4.2	3.7	**3.7**
$35 to $65	4.6	5.3	5.5	**5.2**
$65 to $100	5.3	6.4	6.7	**6.2**
Over $100	5.9	7.2	8.1	**7.2**
Total	**4.4**	**5.8**	**5.1**	**5.2**

Source: Household Diary Study, FY 2005.

Table 4.5:
Transactions Mail Sent by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 35	35 to 54	Over 55	
Under $35	1.0	1.7	1.8	**1.5**
$35 to $65	1.5	2.3	2.5	**2.1**
$65 to $100	2.4	2.4	2.9	**2.5**
Over $100	1.4	2.3	3.6	**2.4**
Total	**1.5**	**2.2**	**2.3**	**2.1**

Source: Household Diary Study, FY 2005.

Household Size

In terms of household size, Tables 4.6 and 4.7 show that going from a one-person household to a two-person household adds about 2.3 pieces of transaction mail per week sent and received, but that further changes have little effect on volume.

Table 4.6:
Transactions Mail Received by Household Size
(Pieces per Household per Week)

Household Size	
One person	3.4
Two	5.2
Three	5.7
Four	5.8
Five or more	6.0
Total	**5.2**

Source: Household Diary Study, FY 2005.

Table 4.7:
Transactions Mail Sent by Household Size
(Pieces per Household per Week)

Household Size	
One person	1.7
Two	2.2
Three	2.1
Four	2.1
Five or more	1.9
Total	**2.1**

Source: Household Diary Study, FY 2005.

For transactions mail received, Table 4.8 shows that each additional adult adds about 1.6 pieces of mail per week, but according to Table 4.9, sent mail only generates .5 pieces of extra mail.

Table 4.8:
Transactions Mail Received by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults in HH	
One	3.5
Two	5.3
Three or more	6.6
Total	**5.2**

Source: Household Diary Study, FY 2005.

Table 4.9:
Transactions Mail Sent by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults in HH	
One	1.7
Two	2.0
Three or more	2.6
Total	**2.1**

Source: Household Diary Study, FY 2005.

Internet Access

Interestingly, Tables 4.10 and 4.11 show that households with Internet access both send and receive more transactions mail, even though the presence of an Internet connection in the home should make that household's transactions more susceptible to electronic diversion. As discussed in Chapter 3, this apparent contradiction is explained in large measure through the fact that household Internet access is strongly correlated with income and education. Additionally, Table 4.11 shows that, while the number of transactions sent by households with Broadband is greater than for households without access, it is also lower than the number for Dial-up users. This is an indication that diversion of bill payments and other transactions is likely to increase as more households upgrade to Broadband.

Table 4.10:
Transactions Mail Received by Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	5.9
Dial-up	5.6
None	3.9
Total	**5.2**

Source: Household Diary Study, FY 2005.

Table 4.11:
Transactions Mail Sent by Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	2.1
Dial-up	2.4
None	1.6
Total	**2.1**

Source: Household Diary Study, FY 2005.

Bill Payment

The average number of bills paid per month per household in 2005 was 12.0, up slightly from 11.5 per month in 2003. Households use a variety of methods to pay bills. Historically, they have been paid in person, via phone, or by mail. In the past decade, emerging technologies provided additional bill receipt and payment options. The most important of these are automatic deduction and on-line bill payment. The Household Diary Study measures bill payment by all of these methods. Table 4.12 shows the percentage of households who

pay bills by each method and the average number of bills paid per month by each method.

Ninety-three percent of households pay at least one of their bills by mail, and the average number paid by mail per household is 8.0 per month, down from 2003 when the average number of bills paid per month by mail was 8.3. Other popular bill payment methods are automatic deduction from bank account (54 percent of households) and in-person (31 percent).

However, the number of actual bills paid by these methods is relatively small (an average of 1.3 and 0.7 pieces per month, respectively).

Notably, households reported paying more monthly bills in 2005 than in 2003 by electronic methods only. As Figure 4.1 shows, these electronic methods account for a growing share of household bill payments over time.

In fact, since 2003, the average number of bills paid by these methods increased by more than 50 percent, largely at the expense of the mail. Figure 4.2 shows that automatic deductions more than doubled since 1998. Given the continuing penetration of the Internet into everyday life, and the increasing affordability and popularity of broadband connections, we expect the trend away from traditional mail to continue in the future.

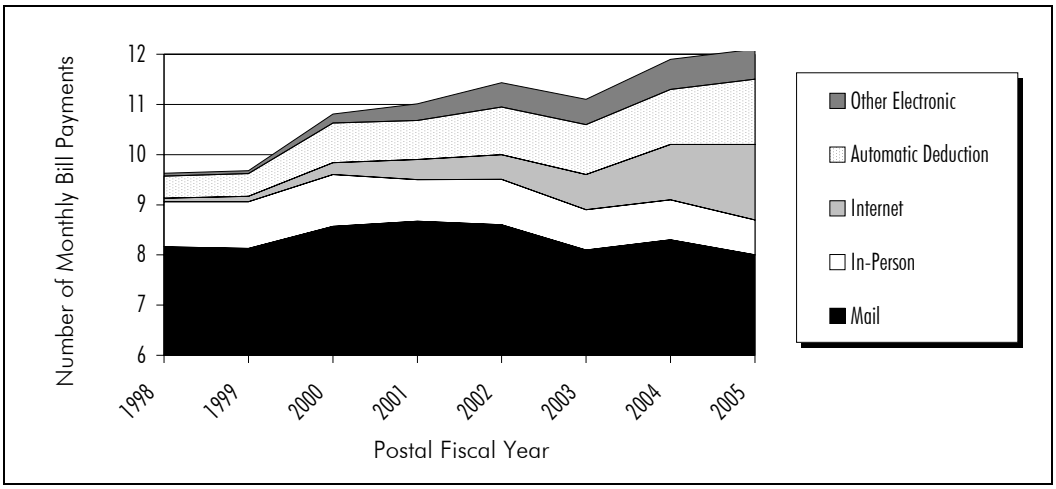
Since 2003, the average number of bills paid by electronic methods increased by more than 50 percent.

Table 4.12:
Bill Payment by Method, FY 2003, 2004, and 2005

Bill Payment Method	2003	2004	2005		
	Average Number of Bills Paid per Month	Average Number of Bills Paid per Month	Average Number of Bills Paid per Month	Share of Bills Paid	Percent of Households Using Method
Mail	8.3	8.3	8.0	67%	93%
Automatic Deduction	1.0	1.1	1.3	11%	54%
Internet	.7	1.1	1.5	12%	28%
In-person	.8	.8	.7	6%	31%
Credit Card	.2	.3	.3	3%	22%
Telephone	.2	.3	.3	2%	14%
ATM	.0	.0	.0	0%	1%
Total	**11.5**	**12.0**	**12.0**	**100%**	**—**

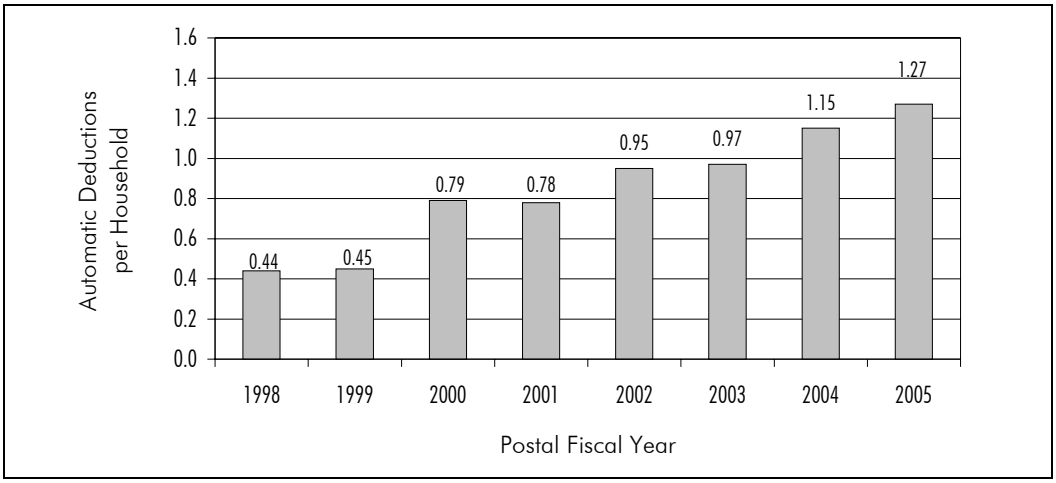
Source: Household Diary Study, FY 2003, 2004 and 2005.

Figure 4.1:
Monthly Average Household Bill Payment by Method



Source: Household Diary Study, FY 1998-2005.

Figure 4.2:
Average Monthly Automatic Deductions per Household



Source: Household Diary Study, FY 1998-2005.

The types of bills paid by mail are shown in Table 4.13. There has been little change in the type of bills and the percentages paid by mail since 2003. Seventy-two percent of households pay their telephone bills by mail, followed by electric bills (68 percent), credit card bills (67 percent), natural gas/other fuel bills (54 percent), and water/sewer bills (49 percent).

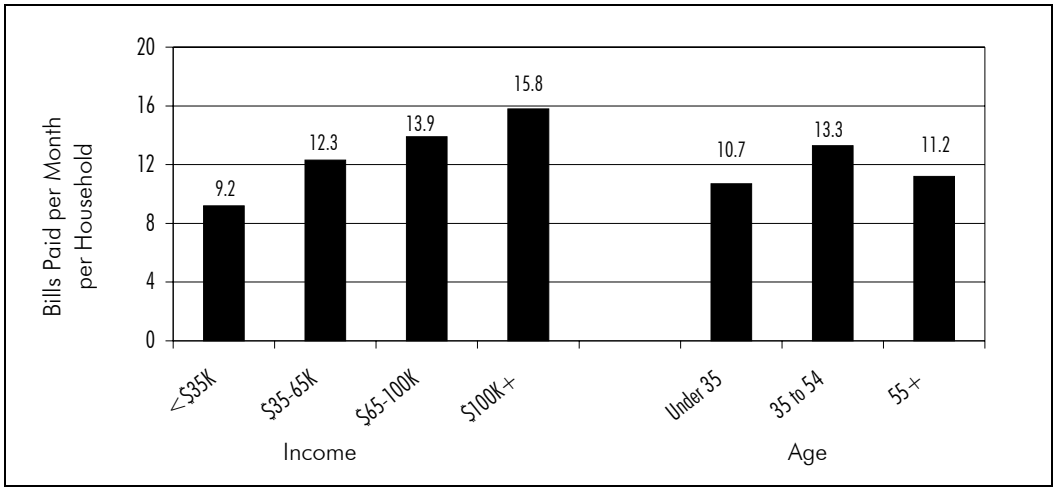
The Household Diary Study finds that the number of total bills paid per month varies by age and income, as does the choice of methods used for bill payment. Figure 4.3 shows the total average number of bills paid per month for each income and age group.

Table 4.13:
Top Bills Paid by Mail

Bill Type	Percent of Household		
	2003	2004	2005
Telephone	72%	71%	72%
Electric	72%	69%	68%
Credit Cards	61%	62%	67%
Natural Gas/ Propane/Fuel	56%	55%	54%
Water/Sewer	51%	50%	49%

Source: Household Diary Study, FY 2003, 2004 and 2005.

Figure 4.3:
Average Bills Paid per Month by Income and Age

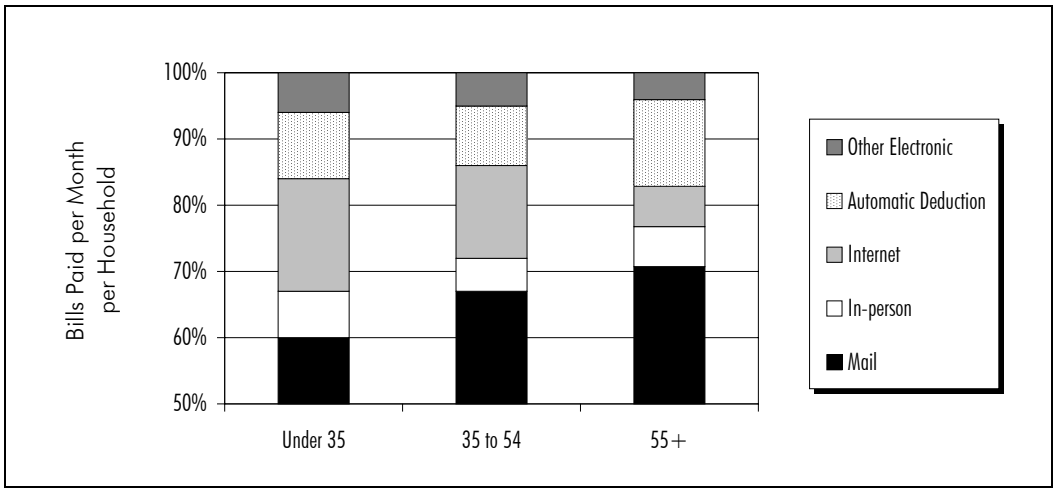


Source: Household Diary Study, FY 2005.

Unsurprisingly, the number of bills paid per month is positively related to household income. Households with incomes over $100,000 paid an average of over fifteen bills per month in FY 2005, compared to the nine bills paid by households with incomes under $35,000. Age has a slightly different relationship with bill payment levels in that younger households (under age 35) and older households (age 55+) pay fewer bills than households where the head of household is between 35 and 54.

More of the bills that younger households pay are paid electronically. As shown in Figure 4.4, the younger the head of a household is, the more likely it is that that household will pay bills electronically. Younger households pay about 60 percent of their bills by mail, and more than 30 percent electronically. Compare that to older households who pay more than 70 percent of bills by mail, and only about 20 percent electronically. The greatest divergence is in Internet bill paying, where younger households pay more than 17 percent of bills using the Internet, while older households pay less than seven percent using this method.

Figure 4.4:
Bill Payment Method by Age



Source: Household Diary Study, FY 2005.

Bills and Statements Received

Table 4.14 shows the overall volume of bills and statements received. The largest volumes of bills originate from credit card companies (4.0 billion), utility companies (2.6 billion), and telephone companies (2.3 billion). Statements received are predominantly sent by the financial sector, including banks, insurance companies, and other financial institutions such as brokerage firms. Overall, the volume of statements received by households from the financial sector increased since 2003. This year, U.S. households received a total of 5.9 billion statements from financial institutions. In 2003, that figure was 5.6 billion.

About 43 percent of First Class Mail received by households is bills and statements. In FY 2005, households received 19 billion bills. This number is about the same as in 2003, but since the number of households increased, this represents a decline in the average number of bills per household. The number of statements received through the mail in FY 2005 increased overall. As shown in Figure 4.5, the increase in the number of statements received by mail was in households with income between $35,000 and $65,000. Households in other income categories received either less or the same number of statements as in 2003.

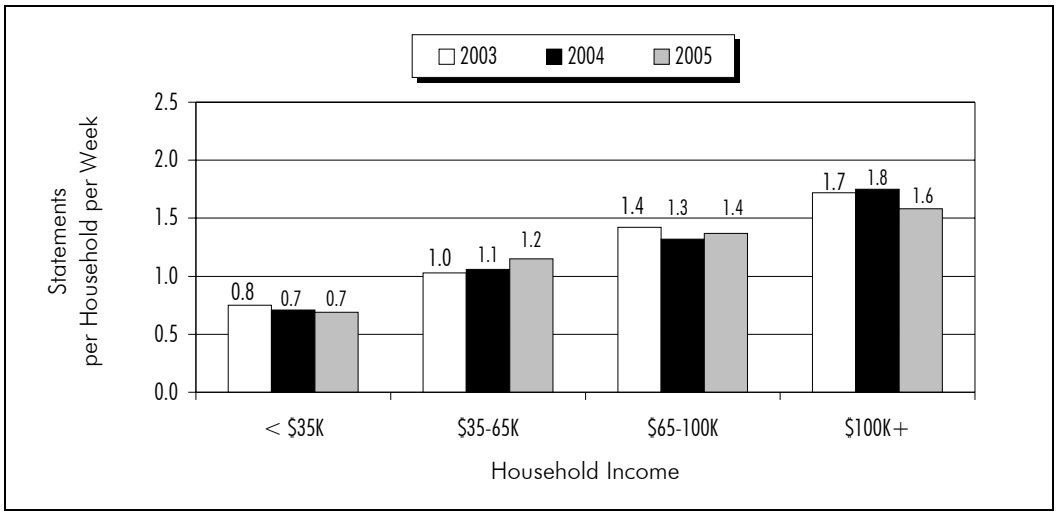
Although the number of bills households receive through the mail is declining on a per household basis, the number of bills received over the Internet is growing rapidly, albeit from a small base. Table 4.15 shows the average volume of bills received by mail and over the Internet.

Table 4.14:
Bill and Statement Volumes by Industry

Industry	Volumes	
	Bills *(Millions)*	**Statements** *(Millions)*
Financial		
Bank, S&L, Credit Union	1,263	3,676
Credit Card	3,981	25
Insurance Company	2,146	497
Real Estate/Mortgage	363	136
Other Financial	139	1,547
Total Financial	**7,892**	**5,881**
Merchants		
Department Store	637	20
Publisher	541	11
Mail Order Company	158	1
Other Merchants	425	18
Total Merchants	**1,761**	**50**
Service		
Telephone Company	2,317	20
Utility Company	2,588	14
Medical and Other Professional	1,914	165
Cable TV	912	9
Other Service	577	45
Total Service	**8,308**	**253**
Manufacturers	58	23
Government	577	295
Social/Nonprofit	8	1
Other/Don't Know/Refused	52	92
Total – All Industries	**18,656**	**6,594**

Source: Household Diary Study, FY 2005.

Figure 4.5:
Statements Received by Mail by Income



Source: Household Diary Study, FY 2003, 2004 and 2005.

Table 4.15:
Average Monthly Bills Received by Method

Method	2003	2004	2005
Mail	13.87	13.76	13.76
Internet	.18	.79	1.02
Total	**14.05**	**14.55**	**14.78**

Source: Household Diary Study, FY 2003, 2004 and 2005.

Chapter 5: Advertising Mail

Introduction

This chapter examines advertising mail, which is any advertising, promotional, or sales material sent through the Postal Service. Advertising mail can be sent as First-Class or Standard Mail.

The Advertising Market

According to McCann-Erickson, American businesses spent about $276 billion in 2005 advertising their products and services, an increase of 4.7 percent over 2004. Of this total advertising spending, nearly 21 percent was spent on direct mail.

Twenty-one percent of total advertising dollars were spent on direct mail advertising in 2005.

Direct mail was the leading media choice of advertisers in 2005, followed by newspapers and television. Direct mail advertising spending increased 8.4 percent from 2004, outpacing the growth in all media categories, except for Internet spending, as shown in Table 5.1.

Table 5.1:
U.S. Advertising Spending by Medium, 2003-2005
(Billions of Dollars)

Medium	2003	2004	2005	Percent Change 2004-2005
Direct Mail	$48.4	$52.2	$56.6	8.4%
Newspapers	$44.8	$46.9	$47.9	2.1%
Television	$41.9	$46.0	$45.3	-1.6%
Radio	$19.1	$19.8	$20.0	0.9%
Magazines	$11.4	$12.1	$12.9	6.1%
Internet	$5.7	$7.1	$7.9	11.6%
All Other	$74.1	$79.5	$85.5	7.5%
Total	**$245.5**	**$263.7**	**$276.0**	**4.7%**

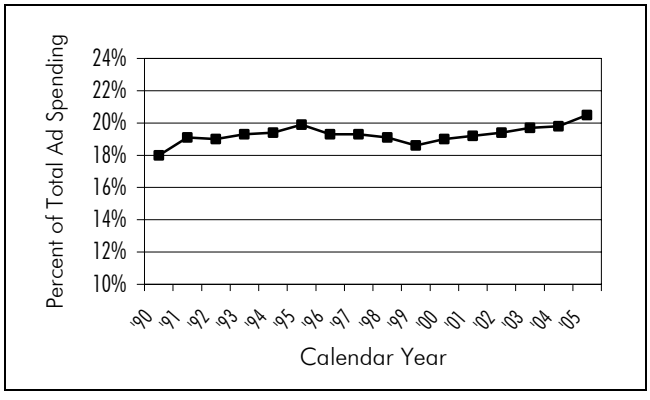
Source: McCann-Erickson, actual for 2003, estimates for 2004 and 2005.
Note: Totals may not add due to rounding.

Despite many changes to the U.S. economy over the past few years, direct mail continues to be the most popular advertising choice. It is a highly efficient and versatile method for communicating with consumers. Direct mail can be targeted to the interests of individual customers. It can be used both to locate new customers and maintain relationships with existing customers. Direct mail allows for a variety of different types of advertising: letters, postcards, catalogs, and free samples. It can be sent as First-Class or Standard Mail, allowing advertisers to trade off their interest in more timely, personalized First-Class mailings against cost savings from Standard Mail.

Importantly, the effectiveness of direct mail is readily measurable, more so than for any other media listed above. Businesses can track the response rate to a mailing far more precisely than for a television commercial or magazine advertisement. This feature alone gives advertising mail a key advantage over its competitors.

Figure 5.1 shows that direct mail's share of total advertising spending has been on a strong upward trend for most of the past 15 years. Since 1999, the direct mail share has risen steadily reaching over 20 percent in 2005. Direct mail has maintained its large ad share even with the introduction of new ad markets such as the Internet.

Figure 5.1:
Direct Mail as a Share of Total Advertising, 1990-2005



Source: U.S. Postal Service calculations based on McCann-Erickson WorldGroup data.

Advertising Mail Volumes

Households received 101.6 billion pieces of advertising mail in 2005, up from 92.0 billion in 2003. Advertising mail represented about 60% of all mail received by households in 2005.

> In 2005, households received 101.6 billion pieces of advertising mail, up from 92 billion in 2003.

Table 5.2 shows that First-Class advertising mail accounts for 18.1 billion pieces, or 18 percent, of all advertising mail received by households. Of this First-Class advertising mail, 10.5 billion pieces are advertising only, while the other 7.5 billion pieces are secondary advertising, such as an advertisement enclosed with a bill. First-Class advertising mail volumes have taken a few hits over the past five years, primarily from diversion to Standard Mail advertising. However, 2005 shows a positive trend with 6 percent growth in volumes over the past three years, putting First-Class advertising mail back on par with 2002 levels. Overall advertising mail to households has grown steadily for the past three years.

On average, U.S. households each receive 17.3 pieces of advertising mail per week; 18.4 percent of that is First-Class advertising. Comparatively, in 2003, U.S. households received an average of 15.9 pieces of advertising mail per week; about 18 percent was First-Class advertising.

About 82 percent of all advertising mail received by households is sent as Standard Mail, a total of 83.5 billion pieces in 2005. This represents a 13 percent increase over 2003.

Over 88 percent of the increase in Standard Mail advertising volumes are found in Regular and ECR mail. In 2005, households received an average of 11.9 Presorted Standard or Bulk Rate mail pieces per week, up from 10.7 in 2003. The exceptional growth in this class of Standard Mail advertising results from constant postal rates for over four years, diversion from First Class Mail advertising, strong growth in Direct Mail advertising spending and a strong overall economy.

> Standard Mail accounts for 82 percent of total advertising mail.

Table 5.2:
Advertising Mail by Mail Classification (Volume in Billions of Pieces)

Mail Classification	Volume *(Billions of Pieces)*			Growth, 2003-2005
	2003	2004	2005	
First-Class Advertising	**17.5**	**16.3**	**18.1**	**3%**
Advertising Only	9.7	8.8	10.5	9%
Secondary Advertising	7.8	7.5	7.5	-4%
Standard Mail	**74.2**	**78.1**	**83.5**	**13%**
Regular and ECR	61.9	65.4	70.1	13%
Nonprofit	12.4	12.8	13.4	8%
Unsolicited Samples (all classes)	.4	.4	.4	-3%
Total Advertising	**92.0**	**94.8**	**101.6**	**10%**
Unaddressed Mail	6.9	6.7	6.9	0%

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Totals may not sum due to rounding. Unaddressed Mail is not included in totals.

Table 5.3: Advertising Mail by Mail Classification (Pieces per Household per Week)

Mail Classification	Pieces per HH per Week			Share of Total
	2003	2004	2005	
First-Class Advertising	**3.0**	**2.8**	**3.1**	**18%**
Advertising Only	1.7	1.5	1.8	10%
Secondary Advertising	1.3	1.3	1.3	7%
Standard Mail	**12.8**	**13.4**	**14.2**	**82%**
Regular and ECR	10.7	11.2	11.9	69%
Nonprofit	2.1	2.2	2.3	13%
Unsolicited Samples (all classes)	.1	.1	.1	0%
Total Advertising	**15.9**	**16.3**	**17.3**	**100%**
Unaddressed Mail	1.2	1.2	1.2	N/A

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Totals may not sum due to rounding. Unaddressed Mail is not included in totals.

Advertising Mail and Household Characteristics

Income, Education, and Age

As advertising mail is used to sell goods and services, it's not surprising that the amount of ad mail received by a household is closely tied to income and education.

The relationship between advertising mail and household income is quite strong, as seen in Table 5.4. Households with less than $35,000 income receive less than half as much advertising mail as households with $100,000 or more income (11.6 pieces per week compared to 26.3). Table 5.4 also shows that education plays a key role in the amount of advertising mail received by households, even after accounting for the impact education has on income.

For example, among households earning under $35,000, ad mail received per week increases as the educational status of the household head increases, rising from 9.9 pieces per week for households headed by someone without a high school degree to 14.6 pieces per week for households headed by a college graduate. The pattern is repeated across nearly all income groups, with more ad mail received as education increases.

The role that education plays in advertising mail is two-fold. First, direct mail is a written communication and education may play some role in its relative effectiveness compared to television or radio advertising. Second, education is not only tied to current household income, but also future household income. A college graduate who currently has a relatively low income may, in a few years, have a much higher income.

Table 5.4: Ad Mail Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Education of Head of Household				Total
	Less than High School	High School graduate	Some College or Technical School	College graduate	
Under $35	9.9	12.2	11.3	14.6	**11.6**
$35 to $65	14.0	16.0	16.9	18.3	**16.6**
$65 to $100	20.4	19.8	20.5	22.3	**20.9**
Over $100	7.4	23.1	25.0	28.4	**26.3**
Total	**12.1**	**15.9**	**16.9**	**21.9**	**17.3**

Source: Household Diary Study, FY 2005.

Table 5.5 shows that households headed by older people receive more advertising mail than those headed by younger people. For every income group, advertising mail received increases as the age of the head of the household increases. In part, this is because age is correlated with other characteristics like marriage, home ownership, and the presence of children in the household.

Moreover, the older a person is, the longer his or her buying history and the more businesses with which the person has had a relationship that advertising mail can help maintain. Households with incomes over $100,000 and with a head of household age 55 and older received the greatest number of advertising mail pieces at 29.7 pieces per week.

The amount of advertising mail received increases as income, education and household size increases.

Table 5.5:
Ad Mail Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 34	35 to 54	Over 55	
Under $35	8.2	10.7	14.0	**11.6**
$35 to $65	13.5	15.7	20.0	**16.6**
$65 to $100	17.9	21.0	23.9	**20.9**
Over $100	21.5	26.2	29.7	**26.3**
Total	**13.1**	**18.3**	**18.8**	**17.3**

Source: Household Diary Study, FY 2005.

Household Size

Tables 5.6 and 5.7 show advertising mail received increases as the size of the household (Table 5.6) and the number of adults in the household (Table 5.7) increase. While this is evident in looking at changes in household size, the greatest change in the number of pieces of advertising mail received is seen in changes in the number of adults in the households.

Table 5.6:
Ad Mail Received by Size of Household
(Pieces per Household per Week)

Household Size	
One person	13.8
Two	18.2
Three	17.7
Four	18.8
Five or more	17.3
Total	**17.3**

Source: Household Diary Study, FY 2005.

As shown in Table 5.7, households with three or more adults receive 56 percent more advertising mail than household with only one adult. This represents an increase from an average of 13.1 pieces per week to 20.5 pieces per week. Note, however, that most of this increase occurs when the number of adults increases from one to two, indicating a strong impact stemming from the likelihood of it being a two-income household.

Table 5.7:
Ad Mail Received by Number of Adults
(Pieces per Household per Week)

Number of Adults	
One	13.1
Two	17.9
Three or more	20.5
Total	**17.3**

Source: Household Diary Study, FY 2005.

Internet Access

Finally, Table 5.8 shows the relationship between advertising mail received and Internet access. Despite all the attention paid to online and e-mail advertising, households with Internet access receive more advertising mail than those without access.

To a large degree, this reflects other characteristics of the household; Internet access is closely tied to income and education. However, it shows that advertising mail continues to be sent even when new advertising media become available. In addition, Table 5.8 may demonstrate the use of Internet information to target potential customers through direct mail advertising.

Table 5.8:
Ad Mail Received by Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	20.7
Dial-up	18.0
None	12.6
Total	**17.3**

Source: Household Diary Study, FY 2005.

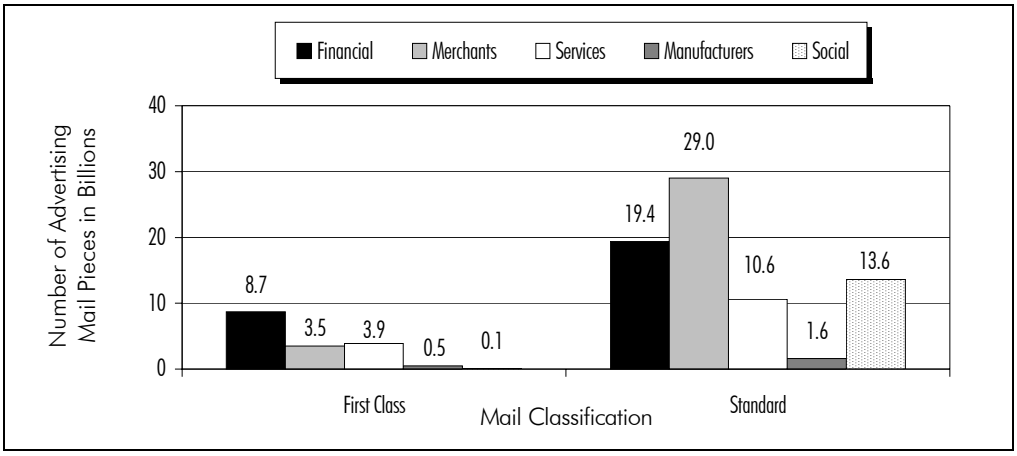
Senders of Advertising Mail

Figure 5.2 provides data on the senders of advertising mail to households. Merchants continue to be the largest senders; in 2005, they sent nearly 40 percent of Standard advertising mail and 21 percent of First-Class advertising mail. Financial firms are the second largest sender and the largest sender of First-Class advertising (52 percent). Many bills and statements, especially those for credit cards, include advertising.

Attitudes Toward Advertising

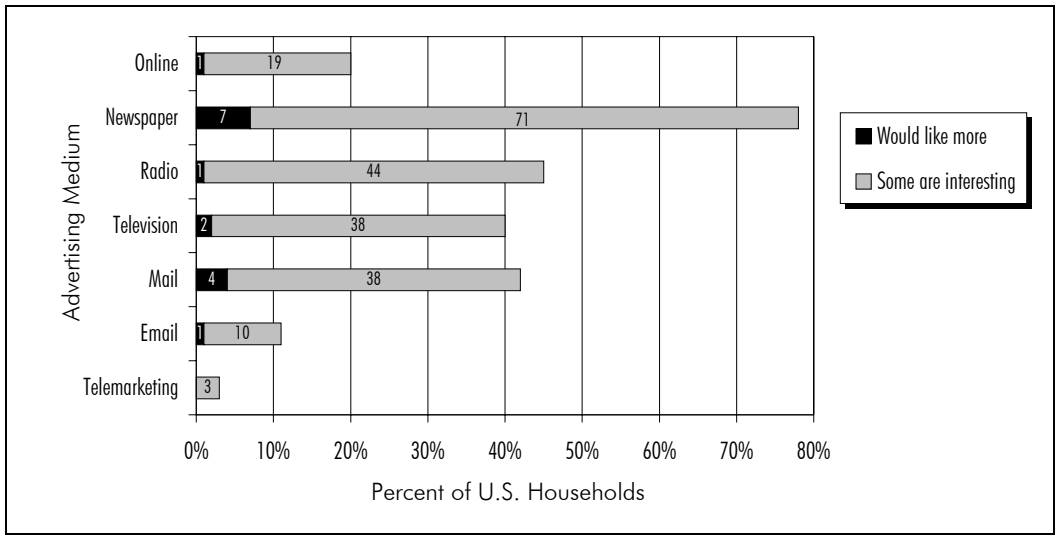
With $276 billion spent in the United States on advertising, it is not surprising that few households wish they received more. Yet many households find some advertising interesting. Figure 5.3 shows about 38 percent of households find some direct mail pieces interesting, a figure that is about the same as for television and radio advertising and significantly less than for newspapers.

Figure 5.2:
Advertising Volumes for First-Class and Standard Mail Advertising by Sender Type



Source: Household Diary Study, FY 2005.
Base: First-Class and Standard Advertising Mail Pieces excluding Unsolicited Samples and pieces for which no industry of sender was given.

Figure 5.3:
Attitudes Toward Advertising by Medium



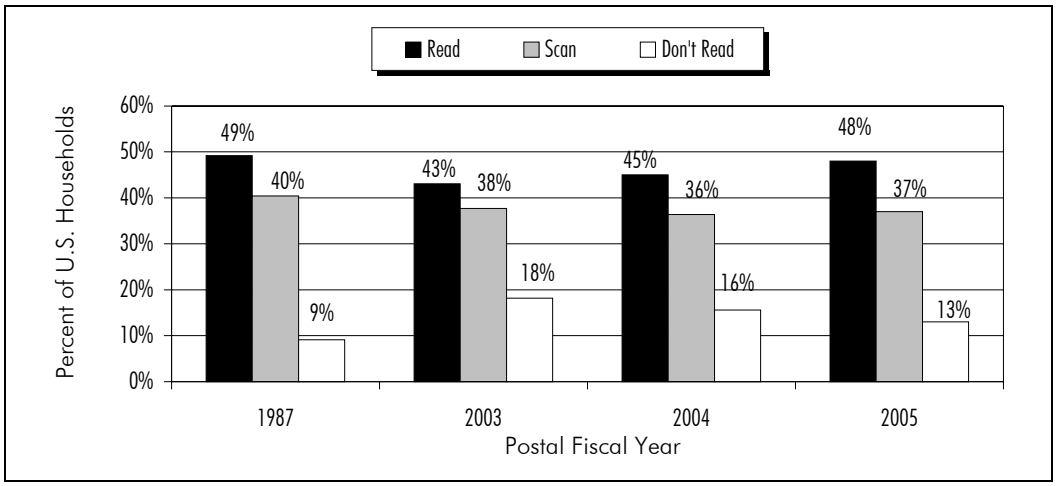
Source: Household Diary Study, FY 2005.
Note: Table does not include households that responded they wished they received less advertising mail or those which provided no response to this question.

The comparison of direct mail with e-mail advertising and telemarketing is particularly revealing. Unlike newspaper, radio or television advertising – where the consumer has chosen to view the media – direct mail, e-mail, and telemarketing are largely unsolicited advertising. Given this, the fact that 38 percent of households find some direct mail advertising interesting is striking when compared with the 10 percent and 3 percent figures for e-mail advertising and telemarketing, respectively.

Whether interesting or not, most households either read or at least scan their advertising mail. Figure 5.4 shows 48 percent of households usually read their advertising mail, while an additional 37 percent scan the mail. Only 13 percent of households report they do not usually read their advertising mail. This represents an increase from the nine percent who did not usually read advertising mail in 1987, but given the large increase in advertising mail volumes since then, it is clear that U.S. households read more advertising mail now than in the past.

Eighty-five percent of households either read or scan advertising mail sent to their household.

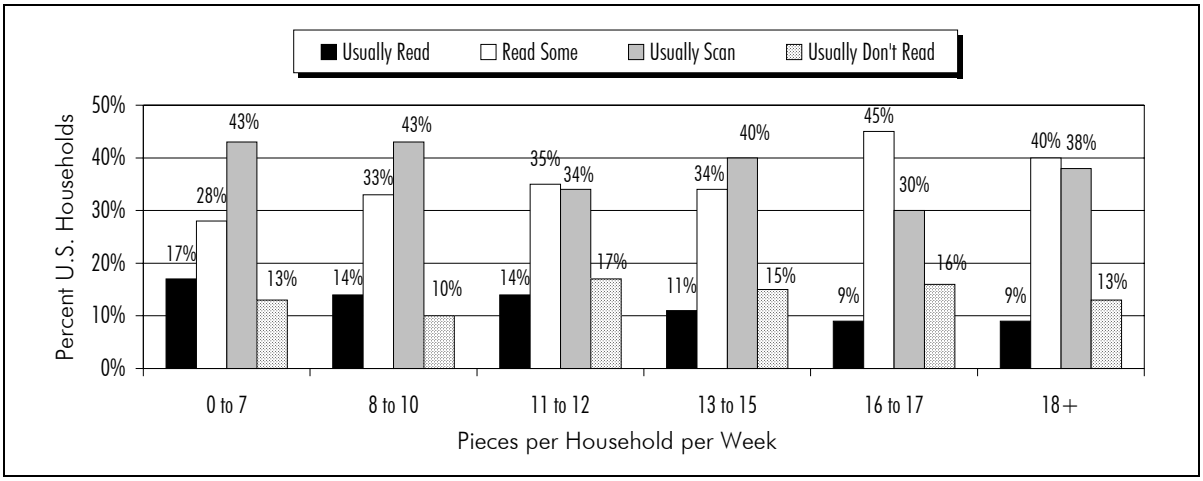
Figure 5.4:
Advertising Mail Behavioral Trends — 1987, 2003, 2004 and 2005



Source: Household Diary Study, FY 1987, 2003, 2004 and 2005.
Note: Percentages do not include those who did not provide a response.

Another interesting result is found in Figure 5.5. Household behavior toward advertising mail is largely independent of how much advertising mail the household receives. For example, among households that receive zero to seven pieces of advertising mail per week, 45 percent usually read all or some of the mail and 13 percent usually do not read any. Among households that receive eighteen or more pieces per week, 49 percent usually read all or some and 13 percent usually do not read any. Thus, households that receive a lot of advertising mail don't appear to be particularly "turned off" by the high volume.

Figure 5.5:
Treatment of Standard Advertising Mail by Number of Standard Mail Pieces Received per Week



Source: Household Diary Study, FY 2005.

Effectiveness of Advertising Mail

Ultimately, advertisers send direct mail because it works – household members read and respond to it. Table 5.9 presents the intended response of households to advertising mail. Households report that they intend to respond to about one in ten pieces of advertising mail, with the intended response to Standard mail being greater than for First-Class advertising (10.1 percent and 8.6 percent of pieces, respectively). The table also shows that households say they may respond to another 16.2 percent of Standard advertising and 9.4 percent of First-Class advertising. This is not to say that a similar mail piece would receive a higher response rate if mailed via Standard Mail; it is more likely the result of a different mix of advertising in Standard Mail. For example, catalogs are routinely mailed Standard Mail, but infrequently mailed First-Class.

Figure 5.6 presents the total number of responses to advertising mail by income. As illustrated in Tables 5.4 and 5.5, higher income households received more advertising mail. Figure 5.6 combines the data on the amount of advertising mail received by household income with the household's intended response to the mail. The result is the average number of responses per week for each income level. For example, households with incomes above $100,000 report they intend to respond to 2.1 pieces of advertising mail per week, and they may respond to another 3.7 pieces per week. Other high-income households also indicate they will respond to more than one piece of advertising mail per week, as do many of the lower income households.

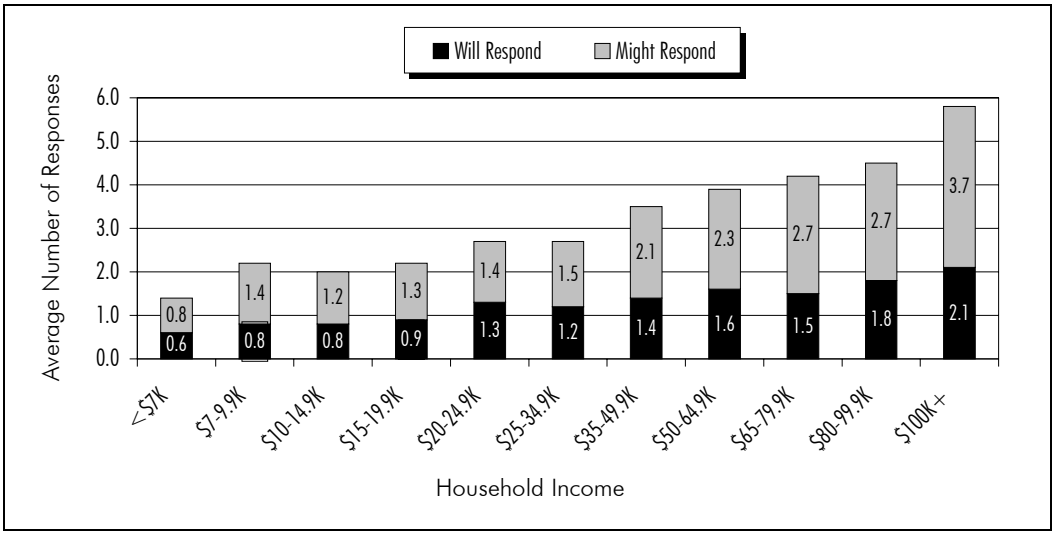
While intended responses do not always lead to actual responses, the data presented in Table 5.9 and Figure 5.6 help explain why direct mail is the number one choice of advertisers in America.

Table 5.9:
Intended Response to Advertising Mail by Class
(Percentage of Pieces)

Response	First-Class	Standard
Yes	8.6%	10.1%
Maybe	9.4%	16.2%
No	66.4%	61.6%
No Answer	15.7%	12.0%

Source: Household Diary Study, FY 2005.

Figure 5.6:
Average Number of Responses by Income



Source: Household Diary Study, FY 2005.

Chapter 6: Periodicals

Introduction

This chapter examines periodicals sent to households. Periodicals Mail consists of newspapers or magazines regularly sent to households, usually as part of a subscription. This chapter analyzes only periodicals delivered by the Postal Service to households. Newspapers or magazines delivered by a local carrier or purchased at a newsstand or grocery store are not included in Household Diary data. The volumes examined here are only a portion of the total periodicals volume since more than one quarter of periodicals received by nonhouseholds go to doctors' offices or other businesses.

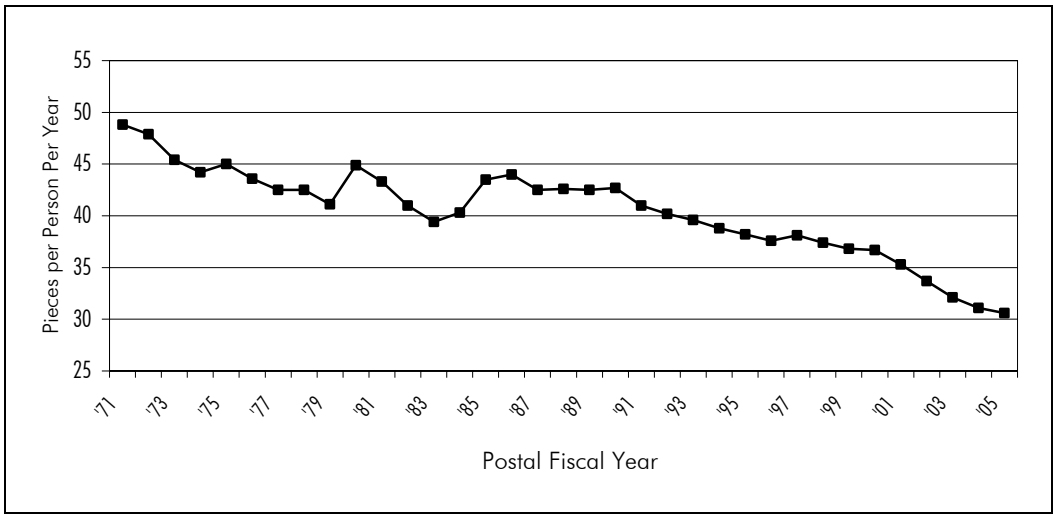
Historically, changes in total Periodicals Mail volumes have not kept pace with population growth.

The Periodical Market

Total Periodicals Mail volumes have not kept pace with population growth over time, as seen in Figure 6.1. Periodicals volumes reached a peak in 1990 at 10.7 billion pieces, but declined each year since 1997, with the exception of FY 2000. In that year, periodicals volumes were temporarily buoyed by an influx of advertising revenue during the dot.com boom. Since FY 2000, the volume dropped dramatically.

Historically, troughs in periodicals volumes have lagged economic slumps more so than with other types of mail, largely because it takes time for some subscription contracts to cycle out. In the most recent recession, periodicals volumes slipped not only because of reduced subscriptions, but also due to the number of publications that ceased operation as sources of advertising revenue dried up.

Figure 6.1:
Periodicals Mail Volume per Person – 1971 to 2005
(Annual Pieces per Person)



Source: U.S. Postal Service, U.S. Census Bureau.

Advertising's Impact on Periodicals

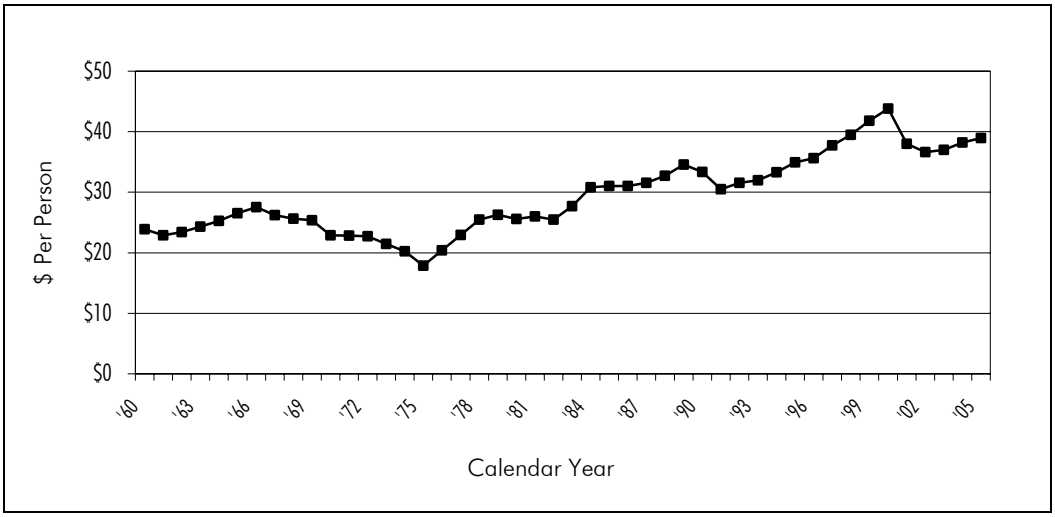
Advertising spending translates into advertising revenue, and the key determinant of magazine profitability is advertising revenue. The trend in advertising spending on magazines shifted to a lower level since its peak in 2000. This suggests that changes in available advertising revenues have altered the profitability of the magazine industry.

Advertising is a form of business investment. As with other investments, when the economy takes a turn for the worse, advertising tends to slow. By 2000, advertising as a percent of Gross Domestic Product (GDP) rose to an historically high level; when the high-tech bubble burst, advertising crashed. In 2001, advertising spending suffered its largest year-over-year decline in history. After three years of economic expansion, overall spending increased; magazine advertising spending also resumed its growth but at a slower pace than the years prior to 2001.

McCann Erickson's projects that advertising spending will continue to grow in 2006, keeping pace with the expected growth in the overall economy. In prior times, this would bode well for new magazine launches as opportunities surface to capture new segments. More magazines in circulation generally translates into higher volume for the Postal Service, since, for most titles, the mail remains the primary distribution channel. More recently, however, the Internet has become a strong competitor of hard-copy publications. The Internet provides an alternative channel for news, information and entertainment. As a consequence, Periodicals volumes may be headed toward long-term decline.

Figure 6.2 shows the sharp decline in real per capita advertising spending for consumer magazines in 2001 and 2002 and the subsequent modest resumption in annual growth, which was slower than the growth in total advertising spending.

Figure 6.2:
Real Per-Capita Consumer Magazine Advertising Spending



Source: McCann Erickson, U.S. Census Bureau.

Household Periodicals Volume

Periodicals represent about 4 percent of all household mail volumes (see Table E-2). Table 6.1 shows the breakdown of periodicals received by households. In FY 2005, households received 6.7 billion periodicals, down about one percent from FY 2003. Seventy-three percent of all periodicals received by households were magazines, a strong increase since 1987, when the share was 59 percent.

In 2005, households received an average of one magazine per week. The most common type of magazine is monthly, accounting for two-thirds of the total magazines.

Newspapers make up 18 percent of total Periodicals volumes, down from the 35 percent share in 1987. The number of newspapers received per household each week declined from 0.6 to 0.2 during that same time period, while magazines only declined from 1.0 to 0.8 pieces per week.

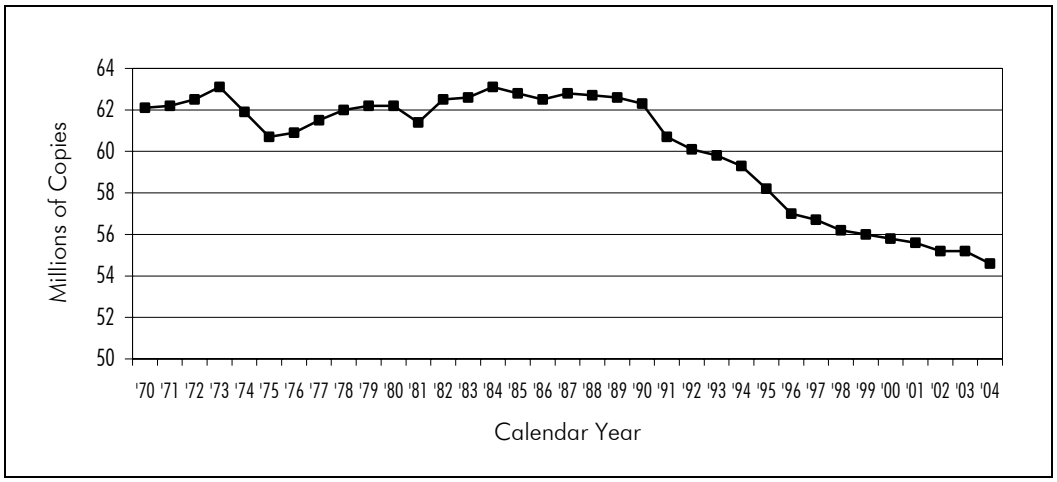
The decline in newspapers captured in the Household Diary study mirrors the behavior seen in overall newspaper circulation. As shown in Figure 6.3, newspaper circulation in general has declined since 1990. Of course, on a per capita basis, the decline is even more pronounced.

Table 6.1:
Periodical Type by Year
(Pieces per Household per Week)

Periodical Type	1987	2004	2005
Newspapers	**.6**	**.3**	**.2**
Daily	.2	.1	.1
Weekly	.3	.1	.1
Other	.1	.1	.0
Magazines	**1.0**	**.8**	**.8**
Weekly	.3	.2	.2
Monthly	.6	.5	.5
Other	.1	.1	.1
Unclassified	**.1**	**.1**	**.1**
Total Periodicals	**1.7**	**1.2**	**1.1**

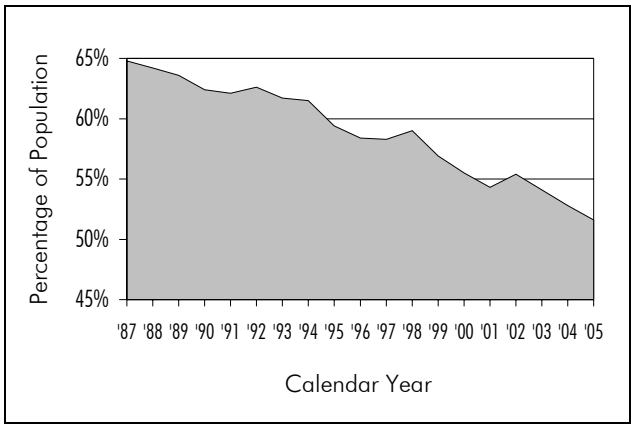
Source: Household Diary Study, FY 1987, 2004, and 2005.
Note: Totals may not sum due to rounding.

Figure 6.3:
Newspaper Circulation



Source: Newspaper Association of America.

Figure 6.4:
Daily Newspaper Readership – 1987 to 2005



Source: Newspaper Association of America.
Note: There was a change in methodology in 1998.

Directly contributing to newspaper volume declines are changes in daily readership levels. As shown above in Figure 6.4, the percentage of the U.S. population reading newspapers on any given day decreased from 65 percent in 1987 to only 52 percent in 2005.

Daily newspaper readership overall has declined since 1987.

Declining newspaper readership and circulation are not the only contributors to the falling volume of newspapers received by households. With current technology and alternate delivery systems, national newspapers such as the *Wall Street Journal*, *The New York Times*, and *USA Today* deliver their papers to prime urban and suburban household customers before breakfast. Local printing/ distribution and morning delivery mean these copies no longer move through the mail.

Periodicals Mail and Household Characteristics

Income, Education, and Age

Table 6.2 shows that as income and education increase, periodicals volume tends to increase. Households where the head householder has a college education receive the most periodicals, averaging 1.5 per week. Similarly, households with incomes over $100,000 get an average of almost two periodicals per week, more than double what households earning less than $35,000 receive.

Households with incomes over $100,000 get more than double the number of periodicals that households earning less than $35,000 receive.

Table 6.3 shows periodicals volume by age and income. The higher the income and age of the household, the higher the volume of periodicals received. For households whose heads are under 34 and with incomes less than $35,000, the average is only 0.5 pieces per week. Households with income over $100,000 and whose heads are over 55 receive the most periodicals.

Table 6.2:
Periodicals by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	.6	.8	.8	.9	**.8**
$35 to $65	.9	1.0	1.0	1.2	**1.1**
$65 to $100	1.6	1.3	1.3	1.5	**1.4**
Over $100	1.6	1.3	1.5	2.0	**1.8**
Total	**.8**	**1.0**	**1.1**	**1.5**	**1.1**

Source: Household Diary Study, FY 2005.

Table 6.3:
Periodicals by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 34	35 to 54	Over 55	
Under $35	.5	.6	1.0	**.8**
$35 to $65	.7	.9	1.5	**1.1**
$65 to $100	.9	1.4	1.8	**1.4**
Over $100	1.4	1.7	2.2	**1.8**
Total	**.7**	**1.1**	**1.4**	**1.1**

Source: Household Diary Study, FY 2005.

Household Size

Table 6.4 and Table 6.5 show that as households increase in size, periodicals volume increases. In households with two adults, periodicals volume is higher than in households with only one, but the presence of additional adults beyond two has no effect on the receipt of periodicals.

Table 6.4:
Periodicals by Size of Household
(Pieces per Household per Week)

Household Size	
One person	1.0
Two	1.3
Three	1.0
Four	1.2
Five or more	1.1
Total	**1.1**

Source: Household Diary Study, FY 2005.

Table 6.5:
Periodicals by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults	
One	.9
Two	1.2
Three or more	1.2
Total	**1.1**

Source: Household Diary Study, FY 2005.

Internet Access

As with many other types of mail, wired households receive more periodicals through the mail as shown in Table 6.6. And, as with those other types of mail, this is largely the result of the correlation between income, education, and Internet access. As household broadband access to the Internet becomes more common, it is likely that more periodical content will be delivered over the Internet, rather than by mail.

Table 6.6:
Periodicals by Type of Internet Access
(Pieces per Household per Week)

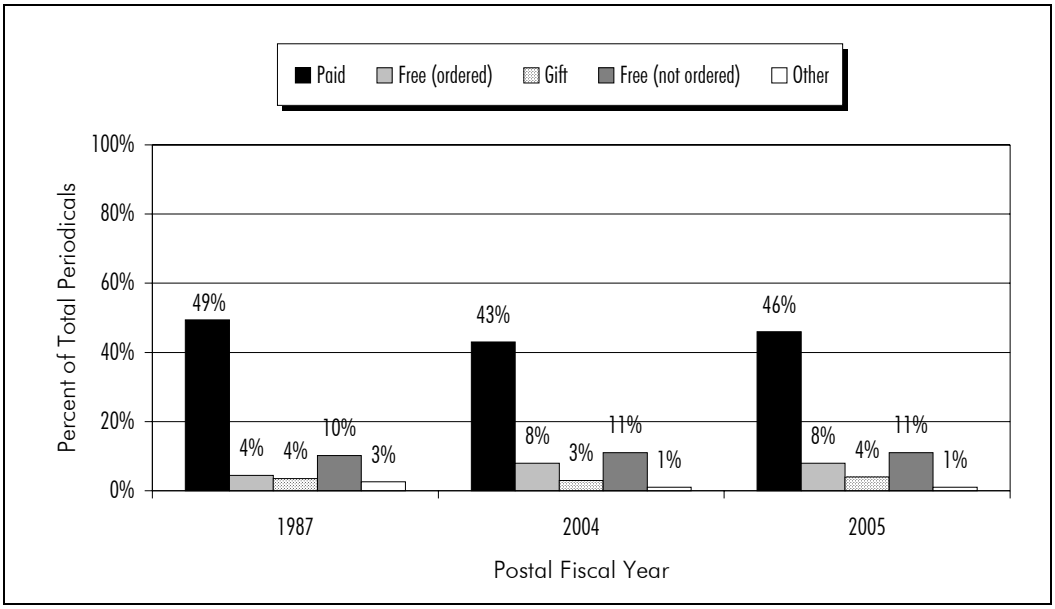
Type of Internet Access	
Broadband	1.3
Dial-up	1.2
None	.9
Total	**1.1**

Source: Household Diary Study, FY 2005.

Subscription Type

Figure 6.5 provides an overview of subscription type for FY 1987, 2004 and 2005. As shown, the distribution of subscription type has remained relatively stable over the 17-year period. In 2005, a household member ordered and paid for 46 percent of total periodicals sent to households. An additional 19 percent were free – either ordered by a household member or delivered to the household without a freestanding order, for example as a prerogative of membership in a professional, fraternal, or religious organization.

Figure 6.5:
Subscription Type by Year



Source: Household Diary Study, FY 1987, 2004 and 2005.
Base: Total Periodicals Mail volume - RPW.
Note: Percentages do not add to 100 due to the exclusion of periodicals sent to non-households and those to which no response was given as to subscription type.

Examining these volumes by sender type shows that commercial rather than member organizations send the vast majority of periodicals. Member organizations are professional affiliations, charitable, religious, and veterans' organizations, educational groups, and unions. As shown in Table 6.7, all member organizations combined account for 25 percent of total periodicals received by households. Charitable organizations account for only one percent of periodicals received by households.

Table 6.7:
Periodicals by Sender Type

Sender Type	Pieces per Household per Week	Percent of Periodicals Received by HH
Commercial Organization	0.85	74%
Professional Organization	0.12	11%
Religious Organization	0.05	5%
Educational Organization	0.06	5%
Union	0.02	2%
Charitable Organization	0.01	1%
Veterans' Organization	0.01	1%
Unclassified	0.03	2%
Total	**1.15**	**100%**

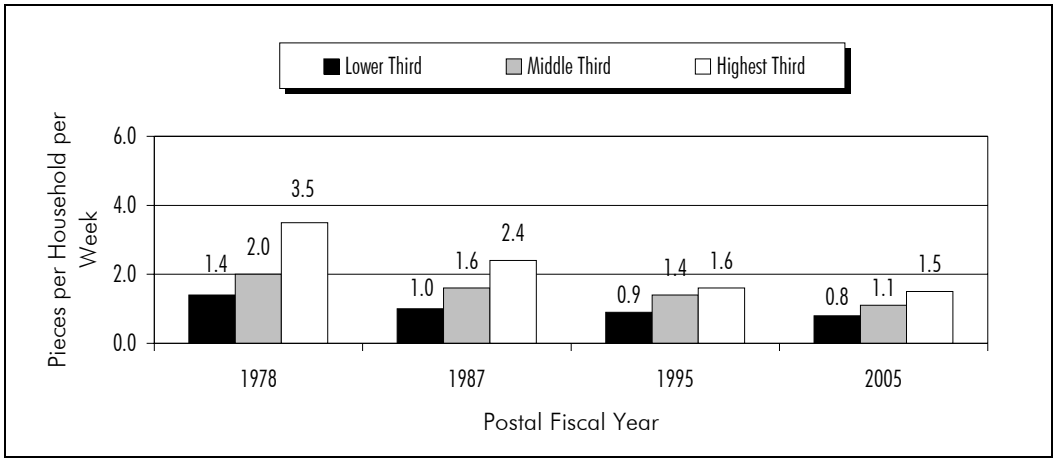
Source: Household Diary Study, FY 2005.

Trends in Readership

A number of factors influence a household's receipt of periodicals. Several of these variables are demographic, while others are more behavioral in nature. In the past, income seemed to influence volume strongly, since periodicals are usually received through a paid subscription.

Typically, higher income households subscribe to more magazines and newspapers. The number of periodicals per household has declined for all income groups, but a new trend has begun to emerge where the decline in pieces per week has been more rapid for high-income households, as shown in Figure 6.6.

Figure 6.6:
Number of Periodicals Received Per Week by Households by Income Group



Source: Household Diary Study, FY 1978, 1987, 1995, and 2005.

Chapter 7: Packages

Introduction

This chapter discusses packages sent and received by households. These packages can be mailed at a variety of rates. Documents are usually sent as First-Class Mail, Priority Mail, or Express Mail. Product samples usually travel as Standard Mail. Merchandise and goods can be any of these classes, or any of the Package Services subclasses, including Parcel Post, Bound Printed Matter and Media Mail.

The Package Market

The package delivery market is an important and growing segment of the economy. From 1996 to 2005, package volume grew from 6.8 billion pieces to 8.1 billion pieces, averaging about 2.0 percent growth per year. Package revenues increased from $37.1 billion to $56.3 billion over the same period, an average of five percent per year. Over that time, however, segments of the market performed differently.

There are three major segments of the package market:

- Overnight air,
- Two- and three-day air, and
- Ground.

The Postal Service is a major player in the two- and three-day air segment but provides services in all: Express Mail in the overnight segment, Priority and First-Class Mail in the two- and three-day segment, and Standard Mail and Package Services in the ground segment.

Table 7.1 shows the volume and growth rates of the three segments of the package delivery market. Between 1996 and 2000, the overall market grew consistently, driven by strong growth in the overnight and two-day and three-day air segments. This growth was largely due to expansion of the services offered by United Parcel Service (UPS) and Federal Express (FedEx). UPS began to push strongly into the overnight market, and both UPS and FedEx developed new two- and three-day offerings to compete with Priority Mail.

The recession in 2001 led to declines in all segments of the package market, with the largest effect in the two-day and three-day air segment as customers shifted to time-guaranteed ground service in the face of hard economic times. The slow recovery in 2002 continued to adversely affect all segments of the package market, except for ground. Ground began to pick up in 2002 at the expense of the other segments. This pattern continued through 2004. As a result, the ground segment accounts for more of the package market by volume than it has at any point over the last ten years (see Figure 7.1). In 2005, the recovering economy boosted all segments of the package market

From 1996 to 2005, package volume grew from 6.8 billion to 8.1 billion pieces.

Many carriers serve the package delivery market; FedEx, UPS, the United States Postal Service, and DHL are the larger players in the market. DHL's recent push in all market segments has intensified competition in what was already a very aggressive market. As seen in Table 7.2, the Postal Service lost market share in every segment in recent years. In part, this is a result of service enhancements made by competitors, such as the introduction of day-certain ground delivery. Competitors also include insurance, tracking and tracing services for all packages as part of the base shipping price, while the Postal Service charges extra fees for these services. As a result, many shippers, especially businesses, view Postal Service package products as inferior to those of competitors.

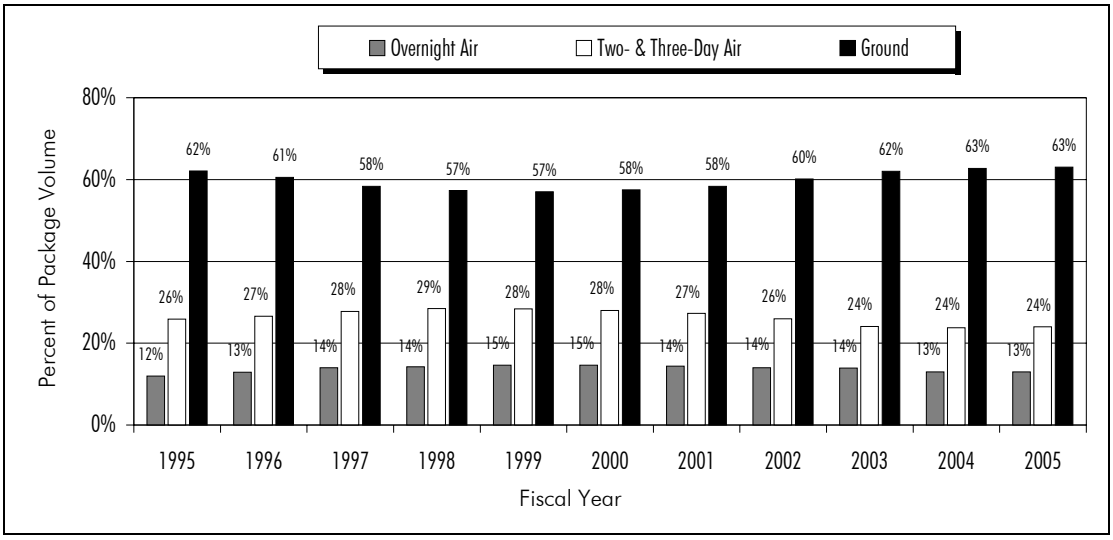
The Postal Service also serves a different niche in the package market than its competitors. Much more of the Postal Service's volume consists of relatively lightweight, low-revenue parcels delivered to households (see Tables 7.3 through 7.5). This is true even in the two- and three-day air segment, where the Postal Service carries the lion's share of the parcels.

Table 7.1:
Total Package Market Volume Growth
(Units in Millions)

Fiscal Year	Overnight Air		Two- & Three-Day Air		Ground		Total	
	Volume	Percent Growth	Volume	Percent Growth	Volume	Percent Growth	Volume	Percent Growth
1996	877	11.5%	1,804	5.8%	4,114	0.7%	6,795	3.3%
1997	985	12.3%	1,962	8.7%	4,112	-0.1%	7,058	3.9%
1998	1,058	7.4%	2,123	8.2%	4,266	3.8%	7,447	5.5%
1999	1,115	5.4%	2,161	1.8%	4,335	1.6%	7,611	2.2%
2000	1,160	4.0%	2,225	3.0%	4,571	5.4%	7,956	4.5%
2001	1,126	-2.9%	2,135	-4.1%	4,555	-0.4%	7,815	-1.8%
2002	1,063	-5.6%	1,980	-7.3%	4,578	0.5%	7,621	-2.5%
2003	1,062	-0.1%	1,842	-7.0%	4,737	3.5%	7,642	0.3%
2004	1,044	-1.7%	1,873	1.7%	4,931	4.1%	7,848	2.7%
2005	1,057	1.2%	1,931	3.1%	5,105	3.5%	8,093	3.1%

Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Figure 7.1:
Package Delivery Market Segment Share



Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Table 7.2:
Postal Service's Volume Market Share

Fiscal Year	Overnight Air	Two- & Three-Day Air	Ground
1995	7%	76%	29%
1996	7%	76%	29%
1997	6%	77%	30%
1998	6%	76%	31%
1999	6%	76%	31%
2000	6%	75%	31%
2001	6%	75%	31%
2002	6%	74%	31%
2003	5%	71%	31%
2004	5%	71%	27%
2005	5%	72%	26%

Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Table 7.3:
FY 2005 Overnight Air Segment Statistics

	Market Share *(Volume)*	Average Revenue per Piece	Weight per Piece *(Pounds)*
FedEx	45%	$16.50	6.5
UPS	29%	$20.40	7.7
DHL	19%	$10.95	4.0
Express Mail	5%	$15.72	0.9
Other	20%	$30.34	10.2

Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Table 7.4:
FY 2005 Two- and Three-Day Air Segment Statistics

	Market Share *(Volume)*	Average Revenue per Piece	Weight per Piece *(Pounds)*
FedEx	13%	$11.51	9.4
UPS	12%	$13.75	10.3
DHL	3%	$8.50	5.8
First-Class & Priority Mail	72%	$3.86	1.3
Other	0%	$22.35	22.1

Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Table 7.5:
FY 2005 Ground Segment Statistics

	Market Share *(Volume)*	Average Revenue per Piece	Weight per Piece *(Pounds)*
FedEx	14%	$6.77	14.6
UPS	56%	$6.65	11.8
DHL	2%	$5.65	10.4
Package Services & Standard Mail	22%	$1.74	2.7
Other	6%	$3.51	2.5

Source: UPS, FedEx and U.S. Postal Service data, Colography Group.

Postal Service Package Volume

Compared to other mail, like letters and flats, the number of packages captured in the Household Diary Study is small, because fewer packages are delivered by the Postal Service. The interpretation of the results should be conducted with this in mind.

Postal Service package volume sent and received by households declined in FY 2005 after increasing in FY 2004. Households sent more packages in 2005 than in 2003, increasing their use of both First-Class and Priority Mail Package Services.

Households received 2.4 billion packages in FY 2005 and sent 541 million, as seen in Table 7.6. When the combined volumes are compared to FY 2003, First-Class and Expedited packages sent by households increased by 13 percent in FY 2005. First-Class and Expedited packages received by households increased by 37 percent compared to FY 2003. Expedited volume has increased over the last three years, due to an improving economy, constant rates and ease of use. The decrease in Package Services and Standard Mail package volume is mostly due to increasing competitive pressure.

Table 7.6:
Postal Service Sent and Received Packages, FY 2003, 2004 and FY 2005
(Units in Millions)

Mail Classification	Volume *(Millions of Pieces)*					
	2003		2004		2005	
	Sent	Received	Sent	Received	Sent	Received
First-Class	198	444	231	508	195	626
Expedited Mail	127	358	211	405	172	477
Standard Mail	—	903	—	887	—	802
Package Services	113	534	178	546	121	399
Unclassified	37	52	60	77	54	71
Total Packages	**475**	**2,291**	**680**	**2,422**	**541**	**2,375**
Mail Classification	**Percent of Pieces**					
	2003		2004		2005	
	Sent	Received	Sent	Received	Sent	Received
First-Class and Priority	42%	19%	34%	21%	36%	26%
Expedited Mail	27%	16%	31%	17%	32%	20%
Standard Mail	—	39%	—	37%	—	34%
Package Services	24%	23%	26%	23%	22%	17%
Unclassified	8%	2%	9%	3%	10%	3%
Total Packages	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Percentages may not sum to 100 due to rounding.
Expedited includes Priority Mail and Express Mail.

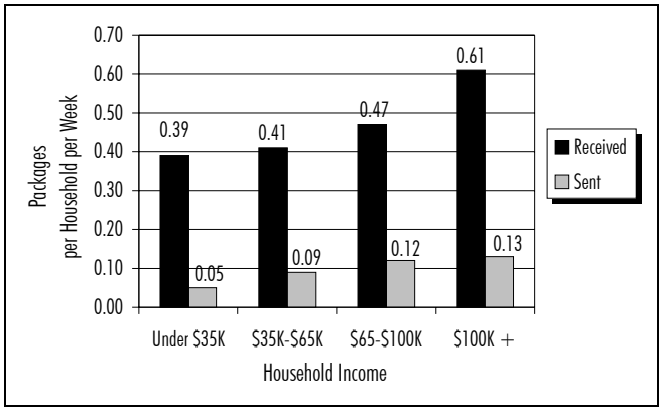
Packages and Household Characteristics

Income, Education, and Age

According to the HDS, high-income households sent and received more packages than their less affluent counterparts in FY 2005, as shown in Figure 7.2.

In fact, households with an income over $100,000 received 56 percent more packages then households with incomes under $35,000. Households headed by younger people on average receive about the same number of packages as those with older heads of household, as shown in Table 7.7. Higher income younger households, however, seem to receive many more packages. This may be the case because young heads of households shop more online than older ones. When it comes to sending packages, lower income younger households send slightly more packages than older households. However, younger households earning over $65,000 send fewer packages, as shown in Table 7.8.

Figure 7.2:
Postal Service Sent and Received Packages by Household Income
(Pieces per Household per Week)



Source: Household Diary Study, FY 2005.
Base: Packages Sent and Received by Households and Delivered by U.S. Postal Service.

Table 7.7:
Postal Service Received Packages by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Total
	Under 35	35 to 54	Over 55	
Under $35	.52	.29	.37	**.39**
$35 to $65	.37	.38	.49	**.41**
$65 to $100	.49	.44	.51	**.47**
Over $100	.81	.59	.53	**.61**
Total	**.48**	**.42**	**.44**	**.44**

Source: Household Diary Study, FY 2005.

Table 7.8:
Postal Service Sent Packages by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of head of Household			Total
	Under 35	35 to 54	Over 55	
Under $35	.06	.05	.04	**.05**
$35 to $65	.14	.07	.07	**.09**
$65 to $100	.04	.15	.13	**.12**
Over $100	.04	.14	.15	**.13**
Total	**.08**	**.10**	**.07**	**.09**

Source: Household Diary Study, FY 2005.

The Household Diary Study indicates that households whose heads have college degrees tend to receive and send more packages on average than households with lower educational attainment. These results are shown in Tables 7.9 and 7.10.

Table 7.9:
Postal Service Received Packages by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Education of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	.45	.35	.37	.43	**.39**
$35 to $65	.20	.48	.39	.49	**.42**
$65 to $100	.29	.47	.45	.50	**.47**
Over $100	.00	.40	.76	.62	**.61**
Total	**.36**	**.43**	**.44**	**.52**	**.44**

Source: Household Diary Study, FY 2005.

Table 7.10:
Postal Service Sent Packages by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Education of Head of Household				Total
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	.02	.03	.05	.16	**.05**
$35 to $65	.00	.12	.07	.12	**.09**
$65 to $100	.00	.20	.09	.11	**.12**
Over $100	.00	.04	.10	.17	**.13**
Total	**.01**	**.10**	**.07**	**.13**	**.09**

Source: Household Diary Study, FY 2005.

Household Size

The Household Diary Study does not show a clear relationship between packages received and household size. Households with more members receive more packages, but larger households do not consistently send more packages, as shown in Tables 7.11 and 7.12. This is probably due to the small sample size and may not be reflective of the true relationship between packages sent and household size.

In FY 2005, households with Internet access sent and received more packages than households without Internet access.

Internet Access

Access to the Internet seems to play an important part in determining the number of packages sent and received by households. Tables 7.13 and 7.14 show the packages sent and received by households with Internet access and households without. In FY 2005, households with broadband Internet access sent four times as many and received 63 percent more packages than households without Internet access. These relationships probably reflect the correlation between income and Internet access.

Table 7.11:
Postal Service Received Packages by Size of Household
(Pieces per Household per Week)

Household Size	
One person	.31
Two	.44
Three	.53
Four	.43
Five or more	.49
Total	**.44**

Source: Household Diary Study, FY 2005.

Table 7.12:
Postal Service Sent Packages by Size of Household
(Pieces per Household per Week)

Household Size	
One person	.04
Two	.09
Three	.11
Four	.08
Five or more	.12
Total	**.09**

Source: Household Diary Study, FY 2005.

Table 7.13:
Received Packages by Household Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	.54
Dial-up	.43
None	.33
Total	**.44**

Source: Household Diary Study, FY 2005.

Table 7.14:
Sent Packages by Household Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	.13
Dial-up	.09
None	.03
Total	**.09**

Source: Household Diary Study, FY 2005.

Household Package Contents

As shown in Table 7.15, packages received by households most often contain music and videos, books, computer hard- and software and clothes, a sign that online purchases and mail order retail are primary drivers of household package volume. In fact, online purchases have contributed to a significant growth in music/video contents received. This type of content increased by more than any other type, growing from 7 percent of all content types in 2003 to 14 percent in 2005.

Households most often send clothes, books, music and videos, and toys. Primarily, this is due to the fact that so many household packages are sent in the holiday season, but it may also be evidence of the growing use of online sales and auction portals, such as eBay.

Table 7.15:
Contents of Postal Service Sent and Received Packages

Contents	Volume *(Millions of Pieces)*					
	2003		2004		2005	
	Sent	Received	Sent	Received	Sent	Received
Music/Video	44	166	114	245	102	452
Clothing	94	274	81	246	75	284
Computer Hard- and Software	18	247	8	259	10	251
Books	61	313	92	319	44	421
Pharmaceuticals/Contacts	4	231	4	178	4	259
Checkbooks	0	127	0	118	1	146
Food Products	13	89	25	128	14	61
Toys	30	85	28	77	24	90
Electronic Equipment	11	79	20	71	14	111
Travel Products and Information	7	27	2	41	8	39
Sporting Goods	5	26	6	29	2	44
Telecommunications	10	9	1	19	5	30
Footwear/Shoes	2	18	3	19	4	31
Other Contents	91	580	139	675	81	983
Total Packages	**390**	**2,271**	**523**	**2,424**	**389**	**3,203**

Table 7.15:
Contents of Postal Service Sent and Received Packages (cont.)

Contents	Percent of Pieces					
	2003		2004		2005	
	Sent	Received	Sent	Received	Sent	Received
Music/Video	11%	7%	22%	10%	26%	14%
Clothing	24%	12%	15%	10%	19%	9%
Computer Hard- and Software	5%	11%	2%	11%	3%	8%
Books	16%	14%	18%	13%	11%	13%
Pharmaceuticals/Contacts	1%	10%	1%	7%	1%	8%
Checkbooks	0%	6%	0%	5%	0%	5%
Food Products	3%	4%	5%	5%	3%	2%
Toys	8%	4%	5%	3%	6%	3%
Electronic Equipment	3%	3%	4%	3%	4%	3%
Travel Products and Information	2%	1%	0%	2%	2%	1%
Sporting Goods	1%	1%	1%	1%	1%	1%
Telecommunications	3%	0%	0%	1%	1%	1%
Footwear/Shoes	1%	1%	1%	1%	1%	1%
Other Contents	23%	26%	27%	28%	21%	31%
Total Packages	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

Source: Household Diary Study, FY 2003, 2004 and 2005.
Note: Expedited Mail sent not included.

Appendix B: Methodology

Study Design and Methodology

The U.S. Postal Service Household Diary Study (HDS), conducted by NuStats on behalf of the Demand Forecasting & Economic Analysis division of the Postal Service's Finance Department, is a continuously fielded study that measures household mail volumes, mail uses and attitudes about the mail and advertising.

The HDS uses a two-stage survey design: Stage 1 is an interviewer-mediated household recruitment interview. Stage 2 is a self-completion mail diary [Appendix C contains the survey instruments]. The HDS uses a multi-mode approach to minimize response bias, improve data accuracy through efficient data checking and householder re-contacts, and provide immediate telephone assistance to participants during their diary week.

Household Recruitment Interview
The **household recruitment interview** collects information on household and personal demographics, recall of mail sent and received, adoption and use of communications technologies, bill payment behavior, and attitudes towards advertising.

Mail Diary
The **mail diary** covers a seven-day period (Monday to Sunday) and collects information on the number of mail pieces received and sent, industry source, mail characteristics, and attitudes regarding mail received.

Sample Design

This section describes the household selection process for participation in the HDS. A sample is the representative subset of the survey population used to gain information about the entire population. The population of inference for the HDS is all U.S. households. The probability design ensures each household has an equal chance of selection.

The sample design allows projections of results to all U.S. households. The Postal Service provided an address sample that NuStats matched for known telephone listings. Generally, the study was conducted using telephone sampling for household selection and screening, followed by diaries mailed to eligible households and completed by each household unit. Households without telephones were contacted via the U.S. Mail. The sample design involves a systematic sample stratified by strata (or urban/rural location) and Census regions, ensuring even coverage across the United States.

A master national sample was specified and drawn by in-house sampling statisticians. The Postal Service drew the household probability sample from the national address database following NuStats specifications. The master list, sorted by zip code, was used to draw a systematic stratified sample. This sample was then tagged with variables indicating each housing unit's geographic location in terms of Census region and stratum.

Sample was drawn for each of the four quarters based on known proportions of households within a Census region and urban or rural location. Census regions are defined by state. Urban and rural location is defined by county and metropolitan status as defined by the U.S. Census Bureau. The strata are defined by county as follows:

- **Stratum 1**: Counties that are part of the thirty largest metropolitan areas in the United States, defined by population, according to Census 2000 100 percent counts.
- **Stratum 2**: Counties that are part of metropolitan areas but are not in Stratum 1.
- **Stratum 3**: Counties that are not part of a metropolitan area.

Quarterly sample frames were then derived based on the amount of sample needed for each quarter, and sample was allocated to region and strata cells based on known proportions as indicated by Census 2000 counts of households.

The sample was continuously "fielded" throughout all 52 weeks of the year. Sample was released in a manner designed to recruit equal sample sizes for each diary week, resulting in a sample file of at least 5,200 households. Table B.1 below shows the distribution of recruited and completed households.

Table B.1:
Sample by Postal Quarter

Quarter	Required Sample	Recruited Households	Completed Households
Quarter 1	1,300	2,149	1,354
Quarter 2	1,300	2,091	1,326
Quarter 3	1,300	2,113	1,317
Quarter 4	1,300	2,138	1,362
Total	**5,200**	**8,491**	**5,359**

Data Collection Method

The study uses a two-stage design in which households are recruited to participate in the diary study in a household interview (Stage 1) and recruited households complete a seven-day diary of mail received and sent (Stage 2).

Stage 1: Household Recruitment Interview

The main function of the household recruitment interview is to recruit households to participate in the diary study. In addition, the interview collects information on household and person demographics, recall of mail sent and received, adoption and use of communication technologies, bill payment behavior and attitudes towards advertising.

Households completed the recruitment interview via computer-assisted telephone interviewing (CATI) technology. The FY 2005 household interview consisted of 8,491 completed interviews with an adult member (age 18 or older) in the household. These respondents represented a cross-section of U.S. households by geography. The household interview contained 155 items and took an average of 22.5 minutes to administer. The flow of the interview included the following elements:

- **Introduction.** As a springboard for detailed questions, each interview began with an introduction, the purpose of the interview, and a battery of questions on how much respondents enjoy certain types of communications activities (e.g., e-mailing, writing cards and letters, reading cards and letters).
- **Mail volume recall.** The respondent was asked to summarize how many personal letters, greeting cards, electronic greeting cards, and packages all members of the household have sent in a particular time period.
- **Use of postal services.** The use of post offices, post office boxes, and private mailing services was explored.
- **Communication with non-US friends and relatives.** If members of the households had any friends or relatives living overseas, they were asked about cards, letters, e-mails, telephone calls, and wire transfers sent.
- **Technology adoption and use.** Questions were asked about ownership and use of personal computers, fax machines, Internet, and other communications technology.
- **Bill payments.** Bill payment volumes, methods, and timing were explored in depth.
- **Periodicals.** Summary volumes of magazines and newspapers received by the household were collected.
- **Advertising.** Descriptions of advertising received by the household as well as attitudes about the advertising, and orders placed because of it, were elicited.
- **Online shopping.** Respondents were asked about their online shopping habits.
- **Financial accounts and credit cards.** Respondents were asked to summarize the total accounts and credit cards held by the household.
- **Household and person demographics.** Demographic items included gender, age, marital status, employment status, educational attainment, race/ethnicity, household income, wage earners in household, home ownership, residence tenure, and dwelling type.

The completion rate for the FY 2005 study (defined as the proportion of respondents who completed the diary portion relative to all recruited respondents) was 63.1 percent. This represents a decrease from 65.7 percent in 2004. Most recruitment refusals took place prior to hearing who NuStats was and why the firm was calling. Refusal households that were later re-contacted cited time constraints and privacy concerns as reasons for not participating.

Stage 2: Mail Diary Package

Recruited households were sent mail diaries, instructions, and a toll-free "help" telephone number. The night before an assigned diary week began, NuStats made reminder calls to households to confirm receipt of the packet and answer any questions. If the packet was not received by this time, NuStats re-confirmed the address, assigned a new diary week, and re-sent the packet.

The diary package contained a Certificate of Appreciation, Instruction Booklet, and a photo-based "Quick Start" sheet. The Instruction Booklet provided information about the study, answers to frequently asked questions, instructions for filling out the diary, guidelines for sorting mail, and examples of mail markings.

The diary instrument was composed of two parts:

- **The Question sheets contained in the Instruction Booklet**. The Question sheets were color-coded by mail classification (e.g. First-Class Mail received, First-Class Mail sent, Standard, Bulk Rate, Nonprofit, etc.). Information collected about each mail classification included: type of mail piece (i.e. envelope, postcard, catalog), receiver zip code, sender zip code, mail classification, mail type, sender type, information about advertising enclosed and receiver reaction or responses to it, and timeliness of the mail piece arrival.
- **Seven answer booklets, each specific to a day of the week**. Each booklet was arranged by mail classification and color-coded to correspond to the question sheets.

Households were instructed to enclose pertinent information from each mail piece received to enable NuStats editors to verify or clarify quantity and classes of mail recorded in the diaries. A three-stage editing process was used to check the accuracy of the diary information recorded by each household. First, returned diary packets were culled for those that represented a reasonable attempt to complete the diary. Second, the diary information recorded for each day was checked to assure sufficient and logical answers as well as to verify recorded information against mail markings returned in the package by the householder. Third, a second editor re-checked the diary information recorded for each day as noted above. This second edit was a quality control check that assured the accuracy of the data. Completed diaries were given a rating of 1 (little follow-up needed), 2 (follow-up call needed), or 3 (unusable).

During the editing process, correction callbacks were made to households to clarify information or to fill-in missing information. About three percent of returned diaries did not pass the edit checking process, while over half required some form of respondent re-contact to clarify or correct diary information.

Of the 8,491 households recruited to receive a diary package, 5,359 actually returned acceptable completed diaries (defined as containing data suitable for analysis) to NuStats for a completion rate of 63.1 percent.

Data Processing

Data Management

Data management entails processing the information resulting from the Household Interview and Mail Diaries, making it available for analysis, storing it and documenting it. Household interviews were conducted using CATI technology, where the questionnaire and relevant data checks were programmed into a master questionnaire. The master survey instrument was then distributed to the individual computer terminals used by interviewers to administer the survey. Recorded data was extracted from the CATI software into a database management file.

Returned diary information was recorded (entered) in one of two methods, either traditional manual data entry or through optical scanning technology. A data entry program, pre-programmed with data check routines, captured approximately 85 percent of the diary data in a database management file. The remaining 15 percent were scanned using Teleform software. Once scanned, the data were translated into a database file that was easily appended to include the manually entered data.

After completion of data collection, editing and entry tasks, the survey data were contained in eleven data files. One data file contained the Household Interview data. The Mail Diary data were in ten files – one for each mail classification (First-Class Mail received, First-Class Mail sent, etc.). These files were all developed in SAS-PC.

The file variables are identified by variable name. For each file variable, the File Information contains:

- Label, which is a brief description of the variable;
- Measurement level, which specifies the level of measurement as scale (numeric data on an interval or ratio scale), ordinal, or nominal. Nominal and ordinal data can be either string (alphanumeric) or numeric;
- Value formats, which identify the response codes; and
- Column width and alignment.

Several SAS programming operations were necessary to put the Mail Diary data in the desired form for analysis. The structure for these programs was contained in a separate File Information document that accompanied the data delivery.

Various edit routines were used to check the consistency of the reported data and to identify reporting or entry errors. Routine edit checks were conducted to examine questionnaire responses for reasonableness and consistency across items. Routine checks included such items as:

- Response code range checks;
- Checks for proper data skips and patterns of answering questions consistent with prior answers;
- Checks for realistic responses (e.g., number of online purchases possible in one month); and
- Checks for high frequency of item non-response (missing data from question refusals).

When conducting these checks, data were compared against the actual survey forms or respondents were re-contacted. NuStats identified extreme values that were impossible or unlikely, and corrected inconsistent data when possible. For example, extremely high numbers of computers owned by a household were examined to determine whether they were legitimate or probable entry errors. Some extreme/inconsistent data values unable to be corrected were edited to missing values.

In addition, NuStats performed in-depth customized data checks to ensure data within each record of the Household Interview were logically consistent. For example, a respondent should report paying bills by Internet only if he/she also reports having Internet access. Customized checks were also used to ensure consistency between the Household Interview and Mail Diary data. For example, an addressee was identified as a child (under 18) in the diary only if the household also reported having a child in the Household Interview.

Raw variables, derived variables, and analytical programs were documented in a data documentation binder that accompanied the data delivery. Any information that could be directly or indirectly used to identify individual respondents, such as respondent names, addresses, or telephone numbers, were removed to protect respondent confidentiality and privacy. Such information is stored in a locked archival file.

Sample Demographic Profile (all counts unweighted), Government Fiscal Year 2005

Table B.2:
Annual Household Income by Recruitment / Retrieval Status

Annual Household Income	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
Under $7,000	50	62	112	1.0	5.3
$7,000 - $9,999	73	90	163	1.5	3.4
$10,000 - $14,999	156	171	327	3.3	6.7
$15,000 - $19,999	217	171	388	4.5	6.3
$20,000 - $24,999	257	173	430	5.4	6.6
$25,000 - $34,999	476	290	766	10.0	11.9
$35,000 - $49,999	672	339	1,011	14.1	14.7
$50,000 - $64,999	820	413	1,233	17.2	12.0
$65,000 - $79,999	694	379	1,073	14.5	9.0
$80,000 - $99,999	570	218	788	11.9	8.3
$100,000 or more	788	429	1,217	16.5	15.8
Don't Know	65	64	129	N/A	N/A
Refused	521	333	854	N/A	N/A
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households that provided a response to the Household Income question.
Population percent based on US Census Bureau, Current Population Survey Annual Demographic File (March 2005).

Table B.3:
Number of Adults in Household by Recruitment / Retrieval Status

Number of Adults in Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
One	1,078	394	1,472	20.1	28.6
Two	3,423	2,115	5,538	63.9	46.1
Three	631	439	1,070	11.8	12.5
Four	185	144	329	3.5	8.2
Five or More	42	40	82	0.8	4.6
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households.
Population percent based on US Census Bureau, Current Population Survey Annual Demographic File (March 2005).

Table B.4:
Geographic Region by Recruitment / Retrieval Status

Geographic Region	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
Northeast	1,055	594	1,649	19.7	19.2
Midwest	1,399	638	2,037	26.1	23.4
South	1,847	1,227	3,074	34.5	36.0
West	1,058	673	1,731	19.7	21.3
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households.
Population percent based on US Census Bureau, Census 2000, Summary File 3, Table H6 (Occupied Housing Units).

Table B.5:
Urban / Rural Location by Recruitment / Retrieval Status

Urban / Rural Location	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
30 Largest Metro Areas	2,587	1,625	4,212	48.3	48.7
Other Metro Areas	1,624	908	2,532	30.3	29.9
Non-Metropolitan Areas	1,148	599	1,747	21.4	21.4
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households.
Population percent based on US Census Bureau, Census 2000; Strata based on Metro Area Classification by County.

Table B.6:
Age of Head of Household by Recruitment / Retrieval Status

Age of Head of Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
18 - 24	75	100	175	1.4	5.7
25 - 44	1,553	1,000	2,553	29.2	37.6
45 - 64	2,357	1,253	3,610	44.3	36.2
65+	1,338	751	2,089	25.1	20.5
Refused	36	28	64	N/A	N/A
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households that provided a valid response.
Population percent based on US Census Bureau, Current Population Survey Annual Demographic File (March 2005).

Table B.7:
Educational Attainment of Head of Household by Recruitment / Retrieval Status

Educational Attainment of Head of Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
8th grade or less	74	121	195	1.4	5.8
Some high school	200	222	422	3.8	9.0
High school graduate	1,547	1,086	2,633	29.0	30.4
Some college	995	547	1,542	18.7	18.4
Technical school graduate	163	96	259	3.1	8.7
College graduate	1,377	659	2,036	25.8	18.0
Postgraduate work	975	362	1,337	18.3	9.7
Refused	28	39	67	N/A	N/A
Total	**5,359**	**3,132**	**8,491**	**100.0**	**100.0**

Note: Sample Percent based only on retrieved households that provided a valid response.
Population percent based on US Census Bureau, Current Population Survey Annual Demographic File (March 2005).

Data Weighting and Expansion

This section explains the methodology used for creating sampling and expansion weights for the FY 2005 Household Diary Study.

The FY 2005 HDS uses both weighting and expansion factors to 1) adjust the sample data to match population parameters and 2) expand mail volumes exhibited in the diary sample to all U.S. households.

Weighting Procedures – FY 2005 Recruitment

Sampling weights were produced separately for the households that participated in the recruitment phase of the FY 2005 HDS, and those that completed and returned a diary. There were two main weighting variables: Geography and Education. FY 2005 recruitment geographic weights were derived from sample households' strata and region:

Strata: As mentioned previously, there are 3 strata. A household was classified within strata as residing in the top 30 metropolitan areas nationwide, any other metropolitan area, or a non-metropolitan area[1]. Table B.8 provides unweighted sample counts from FY 2005 recruitment data for strata:

Table B.8:
HDS 2005 Recruitment Data: Urban/Rural Location

Urban / Rural Location	Household	Percent	Cumulative Percent
30 Largest Metro Areas	4,212	48	48
Other Metro Areas	2,532	30	79
Non-Metro Counties	1,747	21	100%
Total	**8,491**	**100%**	

Regions: Households were classified by state. There are four mutually exclusive regions as defined by the U.S. Census Bureau (along with respective states):

Four Census Regions:

Northeast: Connecticut, Maine, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont.

Midwest: Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota, and Wisconsin.

South: Alabama, Arkansas, Delaware, District of Columbia, Florida, Georgia, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, and West Virginia.

West: Arizona, Alaska, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming.

Table B.9:
HDS 2005 Recruitment Data: Geographic Region

Geographic Region	Households	Percent	Cumulative Percent
Northeast	1,649	20	20
Midwest	2,037	26	46
South	3,074	34	80
West	1,731	20	100%
Total	**8,491**	**100%**	

Strata/Regions: Table B.10 indicates the distribution of households from the FY 2005 recruitment sample within strata and regions.

Population parameters for the intersection of the three strata and four regions were based on 2000 Census counts of households by county. As Table B.10 shows, each county was grouped according to its location within these 12 mutually exclusive and collectively exhaustive geographic categories.

To calculate the weight for each strata/region interval, the population percentage was divided by the sample percentage. Geography weights appear in the last column to the right in Table B.11.

[1] Metropolitan area is defined within the sample according to the official definition used by the U.S. Census Bureau, commonly referred to as Metropolitan Statistical Areas (MSAs). Metropolitan areas are defined as single- or multi-county areas. Non-metropolitan areas are counties that do not belong to a metropolitan area. Each sample county was assigned to a stratum according to its metropolitan status.

Table B.10:
Distribution of Households within Strata and Region

Geographic Region	Stratum *(Urban/Rural Location)*			Total
	30 Largest Metro Areas	Other Metro Areas	Non-Metro Areas	
Northeast	1,133	338	178	**1,649**
Midwest	936	599	502	**2,037**
South	1,050	1,206	818	**3,074**
West	1,093	389	249	**1,731**
Total	**4,212**	**2,532**	**1,747**	**8,491**

Table B.11:
HDS 2005 Recruitment Data: Construction of Geographic Weight

Stratum	Geographic Region	Households *(Population)*	Percent	Households *(Sample)*	Percent	Weight
30 Largest Metro Areas	Northeast	13,512,686	13	1133	13	0.96
	Midwest	11,317,737	11	936	11	0.97
	South	12,488,134	12	1,050	12	0.96
	West	14,020,576	13	1,093	13	1.03
Other Metro Areas	Northeast	4,134,396	4	338	4	0.98
	Midwest	6,617,353	6	599	7	0.89
	South	15,769,481	15	1,206	14	1.05
	West	5,061,183	5	389	5	1.05
Non-Metro Areas	Northeast	2,638,540	3	178	2	1.19
	Midwest	6,799,442	6	502	6	1.09
	South	9,757,599	9	818	10	0.96
	West	3,362,974	3	249	3	1.09
Totals		**105,480,101**	**100%**	**8,491**	**100%**	**1.00**

Source: Household Population Estimates based on U.S. Census Bureau, 2000 Census.

Education: In addition to weighting for differences in geography between the sample and the population, an additional weight was created based on differences in the educational attainment of the head of household. For those households in which either more than one person was identified as the head of household or no individual was identified as the head of household, one was chosen based on the following sequence of criteria: 1) oldest male, 2) oldest female (if no male exists). For cases in which two candidates for the head of the household were of the same age, the respondent on the phone was chosen.

Known population parameters were based on weighted proportions derived from the U.S. Census Bureau's Current Population Survey annual demographic file for March 2005. In cases where the head of household refused to provide his/her education level, an educational level was imputed based on the average educational level of like cases. There were 28 such cases in 2005; mean levels of educational attainment were based on geography (strata and regions), as well as age and income level, if provided.

Table B.12:
HDS 2005 Recruitment Data: Construction of Educational Attainment Weight

Educational Attainment	Households *(Population)*	Percent	Households *(Sample)*	Percent	Weight
8th Grade or Less	6,550,755	6	195	2	2.53
Some high school	10,119,787	9	424	5	1.79
High school graduate	34,278,519	30	2,638	31	0.98
Some college	20,815,588	18	1,574	19	0.99
Technical school graduate	9,871,310	9	286	3	2.59
College graduate	20,318,635	18	2,037	24	0.75
Post graduate work	10,989,456	10	1,337	16	0.62
Totals	**112,944,050**	**100%**	**8,491**	**100%**	**1.00**

Weighting Procedures – FY 2005 Diary Data

As mentioned above, 8,491 households participated in the recruitment phase of the FY 2005 HDS and 5,359 households completed usable diaries. Balancing weights for the FY 2005 HDS diary data were developed in the same way as for the recruitment data. An additional age weight was derived based on the age of the head of household using the following categories: 18-21, 22-24, 25-34, 35-44, 45-54, 55-64, 65-69, and over 70 years old.

Other adjustments to weights used in the diary data included a quarterly adjustment, which accounted for variances in sampling across postal quarters. All component weights were multiplied together and normalized to ensure that the number of weighted cases equals the number of unweighted cases.

A final adjustment in the form of expansion factors was made to expand the sample to the level of total households in the United States at the time of data collection, which was 112.9 million. The number of households in the United States was divided into the number of households that participated in the diary portion of the survey. The resultant factor was applied to each household in the survey. The expansion factor was multiplied by the sampling weight and then multiplied by 52 (the number of calendar weeks in one year) to derive nationwide annual volume estimates from the sample data.

Expansion Factor

112,944,050 / 5,359 = 21,075.58

Component Weight:

$$\omega = \frac{Ps / Pt}{Ss / St},$$

Where Ps = population count in cohort and
Pt = total population count;
Ss = sample count in cohort and
St = total sample count.

Adjustment Factors

In order to account for variations in the reporting of household mail volumes, three types of adjustment factors were used:

1) Destination adjustment factors;

2) Household-to-Household adjustment factors; and

3) Household-to-Nonhousehold adjustment factors.

Destination adjustment factors were calculated from differences between weighted volumes derived from FY 2005 HDS sample data and mailing volumes calculated using the Postal Service's City Carrier Cost System (CCCS) and Rural Carrier Cost System (RCCS). These destination adjustment factors were applied to First-Class Letters & Flats, Bulk Rate & Non-Profit Letters and Flats, Postal Service Packages, and Periodicals.

Household-to-household adjustment factors were applied based on the logic that mail originating and destinating in households form a "closed loop." In other words, mail sent by households to households should equal mail received by households from households. (This situation does not necessarily exist within the confines of a finite sample since households may receive mail from households outside the sampling frame.) Therefore, household mail sent is adjusted to equal household mail received. This factor (1.20) was applied to First-Class Letters & Flats.

Household-to-nonhousehold adjustment factors were applied to account for under-reporting of mail sent by households to nonhouseholds. The use of this adjustment factor is based on a comparison between the reported bills paid by households from the recruitment phase of the survey and amounts derived from actual diary data. This factor (1.21) was applied to First-Class Mail sent by households to non-households.

The following table indicates adjustment factors applied by postal classification:

Table B.13:
HDS 2005 Adjustment Factors Utilized by Postal Classification

Postal Classification	Destination Adjustment Factor	Household-to-Household	Household-to-Nonhousehold
First-Class Letters & Flats	0.94	1.20	1.21
Bulk Rate Letters & Flats	0.99	N/A	N/A
Non-Profit Letters & Flats	0.99	N/A	N/A
Postal Service Packages	1.15	N/A	N/A
Periodicals	0.84	N/A	N/A